PERFECT PAIRING OF THE FIRST D-GROUPS OF A COMPLEX OF TORI AND OF ITS DUAL OVER A CURVE

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Editorial note. The counted claim is the author Proposition duald1 at source lines 1943-1949, and its proof is the proof environment that immediately follows it at source lines 1951-1995; no other proof in the paper is used for it.

Editorial note. The retained core follows the author's Sections 1 to 4 in source order, including the numbered remarks, so the counted statement prints as Proposition 4.12, matching the published article. The corpus record of the same paper's Theorem 4.9 (source label refin2, already counted as problem_id 1933) fixes the section offset used for this reading; the assembled text reproduces that numbering, showing Theorem 4.9 for refin2.

Editorial note. Source lines 1851-1856, a sentence comparing Theorem refin1 with a classical theorem and pointing to the number-field remark cdn2, are not part of this material; that comparison and that remark belong to material outside this card, and no step of the counted proof depends on them.

Editorial note. The inherited layout adapter sources/jep-2020-129-tori-duality/inherited-layout-adapter.tex redefines a command that is already defined under the amsart wrapper and fails to compile; the retained copy sources/jep-2020-129-tori-duality/tori-complex-layout-adapter.tex differs from it only in declaring that command with providecommand, and it is otherwise identical.

Editorial note. The local duality theorem of Demarche used at source line 1986 is an external cited input and is declared as a standard prerequisite; every other step of the counted proof is retained inside this material.

Editorial note. This revision r002 replaces the first assembly of the same counted claim: it adds the numbered items of Section 4 that the published numbering needs and states the resulting number explicitly. No source span of the counted statement or of its own proof changed.

Allowed standard prerequisites (editorial).

- Local duality for two-term complexes of tori over a local field. For a local field complete for a discrete valuation with finite residue field, a two-term complex of tori has perfect local pairings, and the unramified subgroups are mutual annihilators.. Demarche, Theoremes de dualite pour les complexes de tores, Theorems 3.1 and 3.3; scope note sources/jep-2020-129-tori-duality/local-duality-prerequisite-note.md; cited by the author at source line 1986.. Supplies the isomorphism of the ramified vertical map in the counted diagram chase. No global duality result is assumed.
- Classical arithmetic inputs cited inside the retained proofs. Brauer-Hasse-Noether, Hilbert 90, Lang's theorem for tori over finite fields, finiteness of the Brauer group of a finite field, the ideal class group and unit theorem of a global function field, Milne's arithmetic duality theorems III.8.9, II.2.9, II.2.10 and II.3.1, and the local topology of Cesnavicius.. Used only inside

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the retained lemmas and their proofs (bhn, h1lem, fini1, fini2, closedimage, injectcompl, vanishlocal), which are included with their author proofs.

- Homological and topological algebra. The five lemma and the snake lemma, exactness of inverse limits of surjective systems, l-adic and profinite completions, Pontryagin duality for discrete torsion groups, and profinite groups.. The formal mechanism of the completion lemma and of the counted diagram chase.

2. Compact support hypercohomology

Let K be the function field of a smooth, projective, and geometrically integral curve X over a finite field k. Let U be a non empty Zariski open subset of X. Denote by $U^{(1)}$ the set of closed points of U.

Let $\mathscr{C} = (C_p)_{p\in\mathbf{Z}}$ be a bounded complex of fppf sheaves over U. In this text, we define the dual of $\mathscr{C}$ to be the Hom-complex $\widehat{\mathscr{C}}$ defined by

$$\widehat{\mathscr{C}} := \underline{\mathrm{Hom}}^\bullet(\mathscr{C}, \mathbf{G}_m[1]),$$

following the sign conventions in [23, Tag 0A8H] or in [3, X.5.1]. Note that there is a functorial morphism of complexes

$$\mathrm{Tot}(\mathscr{C}\otimes\widehat{\mathscr{C}}) \to \mathbf{G}_m[1]$$

mapping an element $c\otimes\varphi\in C_p\otimes\mathrm{Hom}(C_q, A_n)$ to 0 if $p\neq q$, and to $(-1)^{p(n-1)}\varphi(c)\in A_n$ if $p=q$.

With those conventions, if $\mathscr{C}$ is concentrated in degree 0, i.e., $\mathscr{C}=\mathscr{F}$ with $\mathscr{F}$ an fppf sheaf, then $\widehat{\mathscr{C}}$ is the same as the Cartier dual $\mathscr{F}^D := \underline{\mathrm{Hom}}(\mathscr{F}, \mathbf{G}_m)$ attached in degree -1, i.e., $\widehat{\mathscr{C}} = \mathscr{F}^D[1]$ and the above pairing coincides with the obvious pairing $\mathscr{F}\otimes\mathscr{F}^D[1]\to\mathbf{G}_m[1]$ with no extra sign.

Note also that for any bounded complex $\mathscr{C}$, we have a natural isomorphism of complexes $\widehat{\mathscr{C}[1]} \xrightarrow{\sim} \widehat{\mathscr{C}}[-1]$, given by a sign $(-1)^{n+1}$ in degree n. And given a morphism $f : A\to B$ of bounded complexes, we have a natural isomorphism of complexes $\widehat{\mathrm{Cone}(f)} \xrightarrow{\sim} \mathrm{Cone}(\widehat{f})[-1]$ such that the following diagram commutes

$$\begin{array}{ccccccc} \widehat{A[1]} & \longrightarrow & \widehat{\mathrm{Cone}(f)} & \longrightarrow & \widehat{B} & \xrightarrow{\widehat{f}} & \widehat{A} \\ \downarrow\wr & & \downarrow\wr & & \| & & \| \\ \widehat{A}[-1] & \longrightarrow & \mathrm{Cone}(\widehat{f})[-1] & \longrightarrow & \widehat{B} & \xrightarrow{\widehat{f}} & \widehat{A}. \end{array} \tag{1}$$

If $\mathscr{N}$ is a commutative group scheme (over U or over K), its Cartier dual is denoted $\mathscr{N}^D$. The Pontryagin dual of a topological abelian group A (consisting of continuous homomorphism from A to $\mathbf{Q}/\mathbf{Z}$) is denoted A^*. Unless explicitly specified, the topology used for sheaves (resp. complex of sheaves) and cohomology (resp. hypercohomology) is the fppf topology.

For each closed point v of X, the completion of K at v is denoted by K_v: it is a local field of characteristic p with finite residue field $\mathbf{F}_v$ (observe the slight difference of notation with [8], where K_v stands for the henselization and $\widehat{K_v}$ for the completion). Denote by $\mathscr{O}_v$ the ring of integers of K_v. For every fppf sheaf $\mathscr{F}$ over U with generic fiber F, recall ([8, Prop. 2.1]) the long exact sequence (where the piece of notation $v\notin U$ means that we consider all closed points of $X\setminus U$).

$$\cdots\to H^i_{\mathrm{c}}(U,\mathscr{F})\to H^i(U,\mathscr{F})\to\bigoplus_{v\notin U} H^i(K_v,F)\to H^{i+1}_{\mathrm{c}}(U,\mathscr{F})\to\cdots \tag{2}$$

There is also a long exact sequence

$$\cdots\to H^i_{\mathrm{c}}(U,\mathscr{F}')\to H^i_{\mathrm{c}}(U,\mathscr{F})\to H^i_{\mathrm{c}}(U,\mathscr{F}'')\to H^{i+1}_{\mathrm{c}}(U,\mathscr{F}')\to\cdots \tag{3}$$

associated to every short exact sequence

$$0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$$

of fppf sheaves.

Let us now extend the construction of the groups $H^i_c(U, \dots)$ and [8, Prop. 2.1] to the case of bounded complexes. Let $\mathscr{C} := [\cdots \to \mathscr{F}_i \to \mathscr{F}_{i+1} \to \cdots]$ be a bounded complex of fppf sheaves over U. Let $\mathscr{C} \to I^\bullet(\mathscr{C})$ be an injective resolution of the complex $\mathscr{C}$, in the sense of [23, Tag 013K]. Following [8, §2], let $Z := X \setminus U$ and $Z' := \coprod_{v \in Z} \operatorname{Spec}(K_v) \xrightarrow{i} U$. Denote by $\mathscr{C}_v$ and $I^\bullet(\mathscr{C})_v$ their respective pullbacks to $\operatorname{Spec} K_v$, for $v \notin U$.

We now define $\Gamma_c(U, I^\bullet(\mathscr{C}))$ to be the following object in the category of complexes of abelian groups:

$$\Gamma_c(U, I^\bullet(\mathscr{C})) := \operatorname{Cone}\left(\Gamma(U, I^\bullet(\mathscr{C})) \to \Gamma(Z', i^* I^\bullet(\mathscr{C}))\right)[-1],$$

and $H^r_c(U, \mathscr{C}) := H^r(\Gamma_c(U, I^\bullet(\mathscr{C})))$. We will also denote by $R\Gamma_c(U, \mathscr{C})$ the complex $\Gamma_c(U, I^\bullet(\mathscr{C}))$. Similarly, one can define, for any closed point $v \in X$, complexes $\Gamma_v(\mathscr{O}_v, \mathscr{C})$ computing fppf cohomology groups $H^r_v(\mathscr{O}_v, \mathscr{C})$ over $\operatorname{Spec} \mathscr{O}_v$ with support in the closed point, as in [8, before Lem. 2.6].

As in [8], similar definitions could be made when K is a number field (taking into account the real places), but in this article we will focus on the function field case. However, we will make remarks regularly throughout the text explaining similarities and differences appearing in the number field case.

We will need the analogue of [8, Prop. 2.1 & 2.12] for bounded complexes $\mathscr{C}$: by construction, the first two points of loc. cit., Prop. 2.1 (i.e., exact sequence (2) and (3)) still hold for bounded complexes.

For a complex $\mathscr{C}$ of finite flat group schemes, let us now endow the groups $H^*_c(U, \mathscr{C})$ with a natural topology, compatible with the one defined in [8] in the case where $\mathscr{C}$ is a finite flat group scheme.

Let F be a local field (that is: a field complete for a discrete valuation with finite residue field) and let $\mathscr{C} := [C_r \xrightarrow{f_r} C_{r+1} \to \cdots \to C_s]$ be a bounded complex of finite commutative group schemes over $\operatorname{Spec} F$, with C_i in degree i. We assume that F is of positive characteristic p (if F is p-adic, then all groups $H^r(F, \mathscr{C})$ are finite by [16, Cor. I.2.3]).

Definition 2.1. A morphism $f : G_1 \to G_2$ of topological groups is *strict* if it is continuous, and the restriction $f : G_1 \to f(G_1)$ is an open map (where the topology on $f(G_1)$ is induced by G_2). This is equivalent to saying that f induces an isomorphism of the topological quotient $G_1/\ker f$ with the topological subspace $f(G_1) \subset G_2$.

Let $A := \ker(f_r)$, and let $\overline{\mathscr{C}} := \operatorname{Cone}(A[-r] \xrightarrow{j} \mathscr{C})$. Then there is an exact triangle

$$A[-r] \xrightarrow{j} \mathscr{C} \xrightarrow{i} \overline{\mathscr{C}} \xrightarrow{p} A[1-r]. \tag{4}$$

In addition, we have a natural quasi-isomorphism $\varphi : \overline{\mathscr{C}} \to \mathscr{C}'$, where $\mathscr{C}' := [C_{r+1}/\operatorname{Im}(f_r) \to C_{r+2} \to \cdots \to C_s]$ has a smaller length than $\mathscr{C}$.

There is an alternative dévissage for the complex $\mathscr{C}$, given by the exact triangle:

$$\widetilde{\mathscr{C}} \xrightarrow{i'} \mathscr{C} \xrightarrow{p} C_r[-r] \xrightarrow{\partial} \widetilde{\mathscr{C}}[1], \tag{5}$$

where $\widetilde{\mathscr{C}} := [C_{r+1} \xrightarrow{f_{r+1}} C_{r+2} \to \cdots \to C_s]$.

Recall that for a finite and commutative F-group scheme N, the fppf groups $H^i(F, N)$ are finite if $i \neq 1$ ([16, §III.6]) and they are equipped with a locally compact topology for $i = 1$ by [4]. By induction on the length of $\mathscr{C}$, one deduces

that if $C_i = 0$, then $H^{i+1}(F, \mathscr{C})$ is finite. In particular, with the previous notation, we get that $H^i(F, \mathscr{C})$ is finite if $i \leq r$ or $i \geq s+2$.

We now define a natural topology on $H^i(F, \mathscr{C})$ by induction on the length of $\mathscr{C}$, such that any morphism of such complexes induces a strict map between hypercohomology groups. Using the dévissages given by (4) and (5), one gets the following exact sequences

$$H^{i-1}(F, \mathscr{C}') \to H^{i-r}(F, A) \xrightarrow{f} H^i(F, \mathscr{C}) \xrightarrow{g} H^i(F, \mathscr{C}') \to H^{i+1-r}(F, A) \to \cdots \tag{6}$$

and

$$H^{i-r-1}(F, C_r) \to H^i(F, \widetilde{\mathscr{C}}) \xrightarrow{f'} H^i(F, \mathscr{C}) \xrightarrow{g'} H^{i-r}(F, C_r) \to H^{i+1}(F, \widetilde{\mathscr{C}}) \to \cdots, \tag{7}$$

where all the groups except $H^i(F, \mathscr{C})$ are endowed with a natural topology via the induction hypothesis (observe that $C'_r = 0$).

- Assume that $i = r+1$. Equip $\operatorname{Im} f \cong H^{i-r}(F, A)/\operatorname{Im} H^{i-1}(F, \mathscr{C}')$ with the quotient topology. Then the two rightmost groups in exact sequence (6) are finite, and we can endow $H^{r+1}(F, \mathscr{C})$ with the topology such that $\operatorname{Im} f$ is an open subgroup (see [8, beginning of §3]). In other words it is the finest topology such that f is continuous. Then f and g are strict.
- Assume that $i \neq r+1$. Then $H^{i-r}(F, C_r)$ is finite (and discrete), and using exact sequence (7) one can similarly endow $H^i(F, \mathscr{C})$ with the finest topology making f' continuous. Then both maps f' and g' are strict.

By construction, this topology is functorial in $\mathscr{C}$, i.e., if $\mathscr{C}_1 \to \mathscr{C}_2$ is a morphism of complexes, then the induced map $H^i(F, \mathscr{C}_1) \to H^i(F, \mathscr{C}_2)$ is strict. In addition, given a quasi-isomorphism $\mathscr{C}_1 \to \mathscr{C}_2$, the induced morphism on cohomology groups is a homeomorphism.

Let us now deal with the topology on the groups $H^*_c(U, \mathscr{C})$, where $\mathscr{C}$ is a complex of finite flat commutative group schemes defined over U. Recall that we have an exact sequence analogous to (2):

$$H^{i-1}(U, \mathscr{C}) \to \bigoplus_{v \notin U} H^{i-1}(K_v, \mathscr{C}) \to H^i_c(U, \mathscr{C}) \to H^i(U, \mathscr{C}). \tag{8}$$

We endow the groups $H^i(U,\mathscr{C})$ with the discrete topology, and the groups $H^{i-1}(K_v,\mathscr{C})$ with the topology defined above.

Lemma 2.2. *For all i, the map $H^i(U, \mathscr{C}) \to \bigoplus_{v \notin U} H^i(K_v, \mathscr{C})$ has discrete image.*

Proof. We prove the result by induction on the length of $\mathscr{C}$, using the dévissages induced by the exact triangles (4) and (5).

- Assume that $i = r+1$. Using the exact triangle (5), we get the following commutative diagram of long exact sequences of topological groups (where all maps are strict):

$$\begin{array}{ccccc} H^{r+1}(U, \widetilde{\mathscr{C}}) & \longrightarrow & H^{r+1}(U, \mathscr{C}) & \longrightarrow & H^1(U, C_r) \\ \downarrow & & \downarrow & & \downarrow \\ \bigoplus_{v \notin U} H^{r+1}(K_v, \widetilde{\mathscr{C}}) & \longrightarrow & \bigoplus_{v \notin U} H^{r+1}(K_v, \mathscr{C}) & \longrightarrow & \bigoplus_{v \notin U} H^1(K_v, C_r). \end{array}$$

  The groups on the left hand side are finite, and by [5, Lem. 2.7], the image of the right hand side map is discrete in $\bigoplus_{v \notin U} H^1(K_v, C_r)$. Since $\bigoplus_{v \notin U} H^{r+1}(K_v, \mathscr{C})$ is Hausdorff, an easy topological argument implies that the image of the central vertical map is discrete.

- Assume that $i \neq r+1$. Using the exact triangle (4), we get the following commutative diagram of long exact sequences of topological groups:

$$\begin{array}{ccccc} H^{i-r}(U,A) & \longrightarrow & H^i(U,\mathscr{C}) & \longrightarrow & H^i(U,\mathscr{C}') \\ \downarrow & & \downarrow & & \downarrow \\ \bigoplus_{v\notin U} H^{i-r}(K_v,A) & \longrightarrow & \bigoplus_{v\notin U} H^i(K_v,\mathscr{C}) & \longrightarrow & \bigoplus_{v\notin U} H^i(K_v,\mathscr{C}'). \end{array}$$

  The groups on the left hand side are finite, and by induction on the length of the complex, the image of the right hand side map is discrete in $\bigoplus_{v\notin U} H^1(K_v,\mathscr{C}')$. A similar topological argument as before implies that the central vertical map is discrete. □

As a consequence of this Lemma, one can endow $H^i_{\mathrm{c}}(U,\mathscr{C})$ with the following topology: we put the quotient topology on the group $\bigoplus_{v\notin U} H^i(K_v,\mathscr{C})/\operatorname{Im} H^i(U,\mathscr{C})$ (this topology is Hausdorff), and since $H^i(U,\mathscr{C})$ is discrete, there is a unique topology on $H^i_{\mathrm{c}}(U,\mathscr{C})$ such that the maps in the exact sequence (8) are strict.

Lemma 2.3. *The topological group $H^i_{\mathrm{c}}(U,\mathscr{C})$ is profinite.*

Proof. We prove this Lemma by induction on the length of the complex $\mathscr{C}$. By [8, Prop. 3.5], the lemma is proved when $\mathscr{C}$ is a complex of length one, i.e., concentrated in one given degree.

Given a complex $\mathscr{C}$, consider the previous dévissages:

- Assume that $i = r+1$. Then the exact sequence (6) implies that the group $H^{r+1}_{\mathrm{c}}(U,\mathscr{C})$ is an extension (the maps being strict) of a (discrete) finite group by a profinite group (which is a quotient of $H^1_{\mathrm{c}}(U,A)$ by a closed subgroup), hence $H^{r+1}_{\mathrm{c}}(U,\mathscr{C})$ is profinite.
- Assume that $i \neq r+1$. Then the exact sequence (7) implies that $H^i_{\mathrm{c}}(U,\mathscr{C})$ is an extension of a finite (discrete) group by a profinite group (which is a quotient of $H^i_{\mathrm{c}}(U,\widetilde{\mathscr{C}})$ by a closed subgroup), hence it is profinite. □

3. Cohomology of tori and short complexes of tori

Let U be a non empty Zariski open subset of X. Recall that for every U-*torus* $\mathscr{T}$ (in the sense of [22, IX, Déf. 1.3]), there is a finite étale covering (that can be taken to be connected and Galois) V of U such that $\mathscr{T}_V := \mathscr{T} \times_U V$ is *split*, that is: isomorphic to some power $\mathbf{G}_m^r$ ($r \in \mathbf{N}$) of the multiplicative group ([22, X, Th. 5.16]). The *group of characters* $\widehat{\mathscr{T}}$ of $\mathscr{T}$ is a U-group scheme locally isomorphic to $\mathbf{Z}^r$ for the étale topology, namely it is a torsion-free and finite type $\mathrm{Gal}(V/U)$-module.

Given a complex of U-tori $\mathscr{C} = [\mathscr{T}_1 \xrightarrow{\rho} \mathscr{T}_2]$ with generic fiber $C = [T_1 \xrightarrow{\rho} T_2]$ over K, where by convention $\mathscr{T}_1$ is in degree -1 and $\mathscr{T}_2$ in degree 0, we can apply the construction of Section 2. Namely we have dual complexes $\widehat{\mathscr{C}} = [\widehat{\mathscr{T}_2} \to \widehat{\mathscr{T}_1}]$ and $\widehat{C} = [\widehat{T_2} \xrightarrow{\hat{\rho}} \widehat{T_1}]$ (concentrated in degrees -1 and 0), which are respectively defined over U and over K. Fix a separable closure $\overline{K}$ of K. Denote by S the finite set $X \setminus U$ and by $G_S = \pi_1^{\text{ét}}(U)$ the étale fundamental group of U, which is the Galois group of the maximal field extension $K_S \subset \overline{K}$ of K unramified outside S; then each $\widehat{\mathscr{T}_i}$ ($i = 1, 2$) can be viewed as a discrete G_S-module.

Recall that fppf and étale cohomology coincide for sheaves represented by smooth group schemes ([15, §III.3]) like a torus $\mathscr{T}$, its group of characters $\widehat{\mathscr{T}}$, or finite flat group schemes of order prime to p. In particular (by [15, Lem. III.1.16]) we have for every integer i:

$$\varinjlim_{U} H^i(U,\mathscr{C}) \cong H^i(K,C)$$

(where the limit is over all non empty Zariski open subsets U of X), and likewise for the complex $\widehat{\mathscr{C}}$.

For such 2-term complexes, the pairings of Section 2 can be made explicit (see [7, §2]; note that the sign conventions are slightly different here), and give maps

$$\mathscr{C} \otimes^{\mathbf{L}} \widehat{\mathscr{C}} \to \mathbf{G}_m[1]; \quad C \otimes^{\mathbf{L}} \widehat{C} \to \mathbf{G}_m[1]$$

in the bounded derived category $\mathscr{D}^b(U)$ (resp. $\mathscr{D}^b(\operatorname{Spec} K)$) of fppf sheaves over U (resp. over $\operatorname{Spec} K$). In the case $T_1 = 0$ or $T_2 = 0$, we recover (up to shift) the classical pairings $\mathscr{T} \otimes \widehat{\mathscr{T}} \to \mathbf{G}_m$ and $T \otimes \widehat{T} \to \mathbf{G}_m$ associated to one single torus $\mathscr{T}$. We also have for each positive integer n the n-adic realizations

$$T_{\mathbf{Z}/n}(\mathscr{C}) := H^0(\mathscr{C}[-1] \otimes^{\mathbf{L}} \mathbf{Z}/n); \quad T_{\mathbf{Z}/n}(\widehat{\mathscr{C}}) := H^0(\widehat{\mathscr{C}}[-1] \otimes^{\mathbf{L}} \mathbf{Z}/n)$$

and likewise for C and $\widehat{C}$. The fppf sheaf $T_{\mathbf{Z}/n}(\mathscr{C})$ is representable by a finite group scheme of multiplicative type over U (in the sense of [22, IX, Déf. 1.1]) with Cartier dual $T_{\mathbf{Z}/n}(\widehat{\mathscr{C}})$, and similarly for $T_{\mathbf{Z}/n}(C)$ and $T_{\mathbf{Z}/n}(\widehat{C})$ over K. Besides we have exact triangles ([7, Lem. 2.3]), where for every abelian group (or group scheme) A, the piece of notation ${}_nA$ stands for the n-torsion subgroup of A:

$$ {}_n(\ker \rho)[2] \to \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n \to T_{\mathbf{Z}/n}(\mathscr{C})[1] \to {}_n(\ker \rho)[3] \tag{9}$$

and

$$T_{\mathbf{Z}/n}(\widehat{\mathscr{C}})[1] \to \widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n \to {}_n\widehat{(\ker \rho)} \to T_{\mathbf{Z}/n}(\widehat{\mathscr{C}})[2] \tag{10}$$

in $\mathscr{D}^b(U)$, and similar triangles for C, $\widehat{C}$ in $\mathscr{D}^b(\operatorname{Spec} K)$.

Note also that the objects $\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n$ and $\widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n$ in the derived category have canonical representatives as complexes of fppf sheaves given by

$$\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n = \operatorname{Tot}(\mathscr{C} \otimes [\mathbf{Z} \xrightarrow{n} \mathbf{Z}]) \quad \text{and} \quad \widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n = \operatorname{Tot}(\widehat{\mathscr{C}} \otimes \mathbf{Z}/n).$$

We also have an exact triangle in $\mathscr{D}^b(U)$:

$$(\ker \rho)[1] \to \mathscr{C} \to \operatorname{coker} \rho \to (\ker \rho)[2], \tag{11}$$

where $\operatorname{coker} \rho$ is a torus and $M := \ker \rho$ is a group of multiplicative type, and the dual exact triangle

$$\widehat{(\operatorname{coker} \rho)}[1] \to \widehat{\mathscr{C}} \to \widehat{\ker \rho} \to \widehat{(\operatorname{coker} \rho)}[2]. \tag{12}$$

For every integer i, there are exact sequences

$$\cdots \to H^i(U, \mathscr{T}_1) \to H^i(U, \mathscr{T}_2) \to H^i(U, \mathscr{C}) \to H^{i+1}(U, \mathscr{T}_1) \to \cdots \tag{13}$$

$$\cdots \to H^i(U, \widehat{\mathscr{T}_2}) \to H^i(U, \widehat{\mathscr{T}_1}) \to H^i(U, \widehat{\mathscr{C}}) \to H^{i+1}(U, \widehat{\mathscr{T}_2}) \to \cdots \tag{14}$$

Lemma 3.1. (a) *Let $\mathscr{T}$ be a torus over U. Then $H^0(U, \mathscr{T})$ and $H^1(U, \mathscr{T})$ are of finite type. If $U \neq X$, then $H^1(U, \mathscr{T})$ is finite.*

(b) *Let $\mathscr{N}$ be a finite group scheme of multiplicative type over U. Then $H^1(U, \mathscr{N})$ and $H^2_c(U, \mathscr{N}^D)$ are finite.*

Proof. (a) Let V be an étale and finite connected Galois covering of U such that $\mathscr{T}_V := \mathscr{T} \times_U V$ is isomorphic to $\mathbf{G}_m^r$ for some non negative integer r. The group $\mathscr{T}(V) \simeq H^0(V, \mathbf{G}_m)^r$ is of finite type by Dirichlet's theorem on units. Therefore $H^0(U, \mathscr{T}) \subset H^0(V, \mathscr{T}) = \mathscr{T}(V)$ is also of finite type.

Set $G = \operatorname{Gal}(V/U)$. Then the Hochschild-Serre spectral sequence provides an exact sequence

$$0 \to H^1(G, \mathscr{T}(V)) \to H^1(U, \mathscr{T}) \to H^1(V, \mathscr{T}_V).$$

Since $\mathscr{T}_V \cong \mathbf{G}_m^r$, the group $H^1(V, \mathscr{T}_V) \cong (\operatorname{Pic} V)^r$ is of finite type (resp. finite if $U \neq X$) by finiteness of the ideal class group of a global field. As

$\mathscr{T}(V)$ is of finite type and G is finite, the group $H^1(G, \mathscr{T}(V))$ is finite by [20, Chap. VIII, Cor. 2]. Thus $H^1(U, \mathscr{T})$ is of finite type (resp. finite if $U \neq X$) as well.

(b) We can assume (by ([16, Lem. III.8.9] and [8, Cor. 4.9]) that $U \neq X$. Since every finitely generated Galois module is a quotient of a torsion-free and finitely generated Galois module, the assumption that $\mathscr{N}$ is of multiplicative type implies (by [22, X, Prop. 1.1]) that there is an exact sequence of U-group schemes

$$0 \to \mathscr{N} \to \mathscr{T}_1 \to \mathscr{T}_2 \to 0,$$

where $\mathscr{T}_1$ and $\mathscr{T}_2$ are U-tori. Therefore, there is an exact sequence of abelian groups

$$H^0(U, \mathscr{T}_2) \to H^1(U, \mathscr{N}) \to H^1(U, \mathscr{T}_1).$$

By (a), we know that $H^1(U, \mathscr{T}_1)$ is finite. Let n be the order of $\mathscr{N}$; then the map $H^0(U, \mathscr{T}_2) \to H^1(U, \mathscr{N})$ factorizes through a map $H^0(U, \mathscr{T}_2)/n \to H^1(U, \mathscr{N})$. But $H^0(U, \mathscr{T}_2)/n$ is finite because $H^0(U, \mathscr{T}_2)$ is of finite type by (a). Finally $H^1(U, \mathscr{N})$ is finite. The finiteness of $H^2_c(U, N^D)$ follows by Artin-Mazur-Milne duality ([8, Th. 1.1]). □

Remark 3.2. By dévissage, the finiteness of $H^1(U, \mathscr{N})$ holds for a (not necessarily finite) group of multiplicative type $\mathscr{N}$ because such a group is an extension of a finite group by a torus. Recall also ([16, Lem. III.8.9] and [8, Cor. 4.8]) that for every finite and flat commutative group scheme $\mathscr{N}$ over U, the groups $H^i(U, \mathscr{N})$ and $H^{3-i}_c(U, \mathscr{N})$ are finite if $i \neq 1$ or if $U = X$, and also if p does not divide the order of $\mathscr{N}$ (by [8, Prop. 2.1 (4)] and [16, Th. II.3.1]). Besides these groups are trivial if $i \geq 4$ (this is part of [8, Th. 1.1]).

For an fppf sheaf (or a bounded complex of fppf sheaves) $\mathscr{F}$ on U with generic fiber F, we set (cf. exact sequence (2))

$$D^i(U, \mathscr{F}) = \mathrm{Ker}[H^i(U, \mathscr{F}) \to \bigoplus_{v \notin U} H^i(K_v, F)] = \mathrm{Im}[H^i_c(U, \mathscr{F}) \to H^i(U, \mathscr{F})].$$

Lemma 3.3. *We have $D^2(U, \mathbf{G}_m) = 0$.*

Proof. Let $\mathbf{F}_v$ be the residue field of K_v. By [16, Prop. II.1.1 (b)], we have

$$H^2(\mathscr{O}_v, \mathbf{G}_m) = \mathrm{Br}\, \mathscr{O}_v \cong \mathrm{Br}\, \mathbf{F}_v = 0$$

because $\mathbf{F}_v$ is finite. The Brauer group $\mathrm{Br}\, U$ of U injects into $\mathrm{Br}\, K$ ([15, Cor. IV.2.6]). Now every element of $D^2(U, \mathbf{G}_m) \subset \mathrm{Br}\, U \subset \mathrm{Br}\, K$ has trivial restriction to $\mathrm{Br}\, K_v$ for all places v of K, hence it is trivial by Brauer-Hasse-Noether Theorem ([17, Th. VIII.1.17]). □

Lemma 3.4. *Let $\mathscr{T}$ be a U-torus with generic fiber T.*

(a) *The group $H^1(U, \widehat{\mathscr{T}})$ is finite; the groups $H^0(U, \widehat{\mathscr{T}})$ and $H^0_c(U, \widehat{\mathscr{T}})$ are of finite type and torsion-free. The group $H^1_c(U, \widehat{\mathscr{T}})$ is of finite type.*

(b) *The group $H^2_c(U, \mathscr{T})$ is finite. In particular, if $U = X$, then $H^2(X, \mathscr{T})$ is finite.*

(c) *Assume $U \neq X$. Then $H^2_c(U, \widehat{\mathscr{T}})$ is finite.*

(d) *Assume $i \geq 4$. Then $H^i(U, \widehat{\mathscr{T}}) = H^i_c(U, \widehat{\mathscr{T}}) = 0$. If $U \neq X$, then $H^3(U, \widehat{\mathscr{T}}) = 0$.*

(e) *If $U = X$, then $H^3(U, \widehat{\mathscr{T}}) = H^3(X, \widehat{\mathscr{T}})$ is finite.*

Proof. (a) Let V be a finite connected Galois étale covering of U such that $\mathscr{T} \times_U V$ is split. Let L be the function field of V and set $G = \mathrm{Gal}(L/K)$. We have $H^1(V, \mathbf{Z}) = 0$ because $H^1(V, \mathbf{Z})$ injects into $H^1(L, \mathbf{Z})$ by Leray spectral sequence. Therefore, we have $H^1(V, \widehat{\mathscr{T}}) = 0$, hence the group $H^1(U, \widehat{\mathscr{T}})$ identifies (by the Hochschild-Serre spectral sequence) to a subgroup of $H^1(G, \widehat{T})$, which is finite because $\widehat{T}$ is a G-module of finite type. The assertion about $H^0(U, \widehat{\mathscr{T}})$ (which is a subgroup of $H^0(K, \widehat{T})$) and $H^0_{\mathrm{c}}(U, \widehat{\mathscr{T}}) \subset H^0(U, \widehat{\mathscr{T}})$ is obvious (we even have $H^0_{\mathrm{c}}(U, \widehat{\mathscr{T}}) = 0$ if $U \neq X$). Also the exact sequence

$$\bigoplus_{v \notin U} H^0(K_v, \widehat{T}) \to H^1_{\mathrm{c}}(U, \widehat{\mathscr{T}}) \to H^1(U, \widehat{\mathscr{T}})$$

shows shat $H^1_{\mathrm{c}}(U, \widehat{\mathscr{T}})$ is of finite type.

(b) By (2), there is an exact sequence

$$\bigoplus_{v \notin U} H^1(K_v, T) \to H^2_{\mathrm{c}}(U, \mathscr{T}) \to D^2(U, \mathscr{T}) \to 0.$$

The groups $H^1(K_v, T)$ are finite ([16, Cor. I.2.3]). Since we have $D^2(V, \mathbf{G}_m) = 0$ by Lemma 3.3, a restriction-corestriction argument shows that $D^2(U, \mathscr{T})$ is a subgroup of ${}_nH^2(U, \mathscr{T})$, where $n = \#\,\mathrm{Gal}(V/U)$. By the Kummer sequence in the fppf topology

$$0 \to {}_n\mathscr{T} \to \mathscr{T} \xrightarrow{\cdot n} \mathscr{T} \to 0,$$

the group ${}_nH^2(U, \mathscr{T})$ is a quotient of $H^2(U, {}_n\mathscr{T})$, which is finite (even if p divides n, cf. Remark 3.2). Thus $H^2_{\mathrm{c}}(U, \mathscr{T})$ is finite.

(c) Let $n > 0$. Using the exact sequence in the fppf topology

$$0 \to \widehat{\mathscr{T}} \xrightarrow{\cdot n} \widehat{\mathscr{T}} \to \widehat{\mathscr{T}}/n \to 0, \tag{15}$$

we see that ${}_nH^2_{\mathrm{c}}(U, \widehat{\mathscr{T}})$ is a quotient of $H^1_{\mathrm{c}}(U, \widehat{\mathscr{T}}/n)$, which is finite (see Remark 3.2). It is therefore sufficient to show that $H^2_{\mathrm{c}}(U, \widehat{\mathscr{T}})$ is of finite exponent, and by a restriction-corestriction argument, we reduce to the case $\widehat{\mathscr{T}} = \mathbf{Z}$. As $H^1(K_v, \mathbf{Z}) = 0$, we have

$$H^2_{\mathrm{c}}(U, \mathbf{Z}) = \mathrm{Ker}[H^2(U, \mathbf{Z}) \to \bigoplus_{v \notin U} H^2(K_v, \mathbf{Z})].$$

By [16, Lem. II.2.10], this yields

$$H^2_{\mathrm{c}}(U, \mathbf{Z}) \cong \mathrm{Ker}[H^1(U, \mathbf{Q}/\mathbf{Z}) \to \bigoplus_{v \notin U} H^1(K_v, \mathbf{Q}/\mathbf{Z})],$$

hence

$$H^2_{\mathrm{c}}(U, \mathbf{Z}) \cong \mathrm{Ker}[H^1(G_S, \mathbf{Q}/\mathbf{Z}) \to \bigoplus_{v \notin U} H^1(K_v, \mathbf{Q}/\mathbf{Z})].$$

Therefore, $H^2_{\mathrm{c}}(U, \mathbf{Z})$ is a subgroup of $H^1(\mathrm{Gal}(L/K), \mathbf{Q}/\mathbf{Z})$, where $L \subset K_S$ is the maximal abelian extension of K that is unramified outside S and totally decomposed at every $v \in S = X \setminus U$. The group $\mathrm{Gal}(L/K)$ is isomorphic to $\mathrm{Pic}\, U$ by class field theory, which implies that it is finite because $U \neq X$. Hence $H^2_{\mathrm{c}}(U, \mathbf{Z})$ is finite, which proves (c).

(d) For $i \geq 4$, the groups $H^i_{\mathrm{c}}(U, \widehat{\mathscr{T}})$ and $H^i(U, \widehat{\mathscr{T}})$ coincide thanks to exact sequence (2) because the local field K_v is of strict cohomological dimension 2 ([17, Cor. 7.2.5]). Assume $U \neq X$ and $i \geq 3$. Then, by [16, Prop. II.2.9], we have $H^i(U, \widehat{\mathscr{T}}) = H^i(G_S, \widehat{T})$, which is zero: indeed, G_S is of strict cohomological dimension 2 by [17, Th. 8.3.17]. It remains to deal with the

case $U = X$ (now we assume $i \geq 4$). By [16, Lem. II.2.10], the group $H^i(X, \mathbf{Z})$ is torsion; by a restriction-corestriction argument, the same holds for $H^i(X, \widehat{\mathscr{T}})$. Since $\mathbf{Q}$ is uniquely divisible, this yields

$$H^i(X, \widehat{\mathscr{T}}) = H^{i-1}(X, \widehat{\mathscr{T}} \otimes \mathbf{Q}/\mathbf{Z}) = \varinjlim_n H^{i-1}(X, \widehat{\mathscr{T}}/n).$$

For $i \geq 5$, the group $H^{i-1}(X, \widehat{\mathscr{T}}/n)$ is zero (cf. Remark 3.2), so we are done. Assume $i = 4$. We observe that the finite group $H^3(X, \widehat{\mathscr{T}}/n)$ is dual to $H^0(X, {}_n\mathscr{T})$ by Artin-Mazur-Milne duality ([8, Th. 1.1]), so the dual of the discrete torsion group $H^4(X, \widehat{\mathscr{T}})$ is the profinite group

$$\varprojlim_n H^0(X, {}_n\mathscr{T}) = \varprojlim_n {}_n(T(K))$$

(the equality holds because the X-group scheme ${}_n\mathscr{T}$ is finite and X is connected). But K is a global field, hence $T(K)_{\mathrm{tors}}$ is finite: indeed, if L is a finite extension of K such that T splits over L, then $T(K) \subset T(L)$ with $T(L) \simeq (L^*)^r$ for some r, and L^* contains only finitely many roots of unity. Therefore, $T(K)$ has trivial Tate module, which yields the result.

(e) Using exact sequence (15), we get a surjection $H^2(X, \widehat{\mathscr{T}}/n) \to {}_nH^3(X, \widehat{\mathscr{T}})$, so it is sufficient (by Remark 3.2) to show that $H^3(X, \widehat{\mathscr{T}})$ is of finite exponent. By restriction-corestriction, we can therefore assume that $\widehat{\mathscr{T}} = \mathbf{Z}$. By the same method as in (d), we get that the dual of $H^3(X, \mathbf{Z})$ is $\varprojlim_n H^1(X, \mu_n)$. As $H^0(X, \mathbf{G}_m) = k^*$ because X is a proper and geometrically integral curve, we get an exact sequence of finite groups

$$0 \to k^*/k^{*n} \to H^1(X, \mu_n) \to {}_n \operatorname{Pic} X \to 0.$$

Since $\operatorname{Pic} X$ is of finite type, we have $\varprojlim_n ({}_n\operatorname{Pic} X) = 0$, hence $\varprojlim_n H^1(X, \mu_n)$ is the inverse limit of the k^*/k^{*n}, which is k^* itself because k is finite. Thus $H^3(X, \mathbf{Z})$ is the dual of k^*, which is finite (but not zero). □

Remark 3.5. Assume $U = X$. Then the group $H^2_{\mathrm{c}}(U, \widehat{\mathscr{T}}) = H^2(X, \widehat{\mathscr{T}})$ is in general infinite: for example $H^2(X, \mathbf{Z}) \cong H^1(X, \mathbf{Q}/\mathbf{Z})$ is the dual of the étale fundamental group $\pi_1^{\text{ét}}(X)$ and the latter is an extension of $\mathrm{Gal}(\bar{k}/k) = \widehat{\mathbf{Z}}$; therefore $H^2(X, \mathbf{Z})$ contains a copy of $\mathbf{Q}/\mathbf{Z}$.

Proposition 3.6. *Let $\mathscr{C} = [\mathscr{T}_1 \xrightarrow{\rho} \mathscr{T}_2]$ be a complex of U-tori with generic fiber $C = [T_1 \to T_2]$.*

(a) *Let $i \in \{-1, 0\}$. Then the groups $H^i(U, \widehat{\mathscr{C}})$ and $H^i_{\mathrm{c}}(U, \widehat{\mathscr{C}})$ are of finite type, and the restriction map $H^i(U, \widehat{\mathscr{C}}) \to H^i(K, \widehat{C})$ is an isomorphism. The restriction map $H^1(U, \widehat{\mathscr{C}}) \to H^1(K, \widehat{C})$ is injective. If $U \neq X$, then $H^1_{\mathrm{c}}(U, \widehat{\mathscr{C}})$ is of finite type.*

(b) *The groups $H^{-1}(U, \mathscr{C})$ and $H^{-1}_{\mathrm{c}}(U, \mathscr{C})$ are of finite type, and so is $H^0(U, \mathscr{C})$. If $U = X$, then $H^1(U, \mathscr{C}) = H^1(X, \mathscr{C})$ is of finite type.*

(c) *Assume $U \neq X$. Then $D^1(U, \mathscr{C})$ and $D^1(U, \widehat{\mathscr{C}})$ are finite.*

Proof. (a) The fact that $H^i(U, \widehat{\mathscr{C}})$ and $H^i_{\mathrm{c}}(U, \widehat{\mathscr{C}})$ are of finite type for $i \in \{-1, 0\}$ follows immediately by dévissage (cf. exact sequences (14) and (2)) from Lemma 3.4 (a), and we even have $H^{-1}_{\mathrm{c}}(U, \widehat{\mathscr{C}}) = 0$ if $U \neq X$. For a U-torus $\mathscr{T}$, the restriction map $H^0(U, \widehat{\mathscr{T}}) \to H^0(K, \widehat{T})$ obviously is an isomorphism. Let V be a connected Galois covering of U with function field L and group G, such that $\mathscr{T}$ splits over V. As seen before, we

have $H^1(V,\mathbf{Z})=H^1(L,\mathbf{Z})=0$, hence $H^1(V,\widehat{\mathscr{T}})=H^1(L,\widehat{T})=0$. By the Hochschild-Serre spectral sequence we get $H^1(U,\widehat{\mathscr{T}})\cong H^1(K,\widehat{T})$ because both groups identify to $H^1(G,\widehat{T})$. By [16, Lem. II.2.10], we have

$$H^2(U,\widehat{\mathscr{T}})\cong H^1(U,\widehat{\mathscr{T}}\otimes\mathbf{Q}/\mathbf{Z})=\varinjlim_{n>0}H^1(U,\widehat{\mathscr{T}}/n),$$

and $H^1(U,\widehat{\mathscr{T}}/n)\hookrightarrow H^1(K,\widehat{T}/n)$ because $\widehat{\mathscr{T}}/n$ is a finite U-group scheme, which implies $H^2(U,\widehat{\mathscr{T}})\hookrightarrow H^2(K,\widehat{T})$.

The commutative diagram with exact lines

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & H^i(U,\widehat{\mathscr{T}_1}) & \longrightarrow & H^i(U,\widehat{\mathscr{T}_2}) & \longrightarrow & H^i(U,\widehat{\mathscr{C}}) & \longrightarrow & H^{i+1}(U,\widehat{\mathscr{T}_1}) & \longrightarrow & H^{i+1}(U,\widehat{\mathscr{T}_2}) \\
 & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
0 & \longrightarrow & H^i(K,\widehat{\mathscr{T}_1}) & \longrightarrow & H^i(K,\widehat{T_2}) & \longrightarrow & H^i(K,\widehat{C}) & \longrightarrow & H^{i+1}(K,\widehat{T_1}) & \longrightarrow & H^{i+1}(K,\widehat{T_2})
\end{array}$$

and the five lemma now give that the restriction map $H^i(U,\widehat{\mathscr{C}})\to H^i(K,\widehat{C})$ is an isomorphism for $i\in\{-1,0\}$, and is injective for $i=1$. The fact that $H^1_c(U,\widehat{\mathscr{C}})$ is of finite type if $U\neq X$ is immediate by dévissage thanks to Lemma 3.4 (c).

(b) The first two assertions follow from Lemma 3.1, using exact sequence (13). For $U=X$, every X-torus T satisfies that $H^1(X,\mathscr{T})$ is of finite type (Lemma 3.1) and $H^2(X,\mathscr{T})$ is finite (Lemma 3.4 (b), whence the result.

(c) By functoriality, the image of $D^1(U,\mathscr{C})$ by the map $u: H^1(U,\mathscr{C})\to H^2(U,\mathscr{T}_1)$ is a subgroup of $D^2(U,\mathscr{T}_1)$. The latter is finite by Lemma 3.4 (b), because it is a quotient of $H^2_c(U,\mathscr{T}_1)$. As the kernel of u is a quotient of $H^1(U,\mathscr{T}_2)$ (which is finite by Lemma 3.4 (a), this means that $D^1(U,\mathscr{C})$ is finite.

The group $H^2_c(U,\widehat{\mathscr{T}_2})$ is finite by Lemma 3.4 (c). Hence $D^2(U,\widehat{\mathscr{T}_2})$ is finite. On the other hand, the kernel of the map $H^1(U,\widehat{\mathscr{C}})\to H^2(U,\widehat{\mathscr{T}_2})$ is a quotient of the group $H^1(U,\widehat{\mathscr{T}_1})$, which is finite by Lemma 3.4 (a). Thus $D^1(U,\widehat{\mathscr{C}})$ is finite. □

Lemma 3.7. *The image I of the group $H^0(U,\mathscr{C})$ into $\bigoplus_{v\notin U}H^0(K_v,C)$ is a discrete (hence closed) subgroup, and so is the image of $H^{-1}(U,\mathscr{C})$ into $\bigoplus_{v\notin U}H^{-1}(K_v,C)$.*

Proof. We can assume $U\neq X$. Let us start with the case when $\mathscr{C}=\mathbf{G}_m$. Then $\mathscr{O}_U^*:=H^0(U,\mathbf{G}_m)$ is a discrete subgroup of $\bigoplus_{v\notin U}K_v^*$, because its intersection with the open subgroup $\bigoplus_{v\notin U}\mathscr{O}_v^*$ is $H^0(X,\mathbf{G}_m)=k^*$, which is finite. Consider now a U-torus $\mathscr{T}$. Let W be a connected Galois finite covering of U (with function field $L\supset K$) that splits $\mathscr{T}$. Let $G:=\mathrm{Gal}(L/K)$. By the case $\mathscr{T}=\mathbf{G}_m$, the subgroup $H^0(W,\mathscr{T})$ is discrete in $\bigoplus_{w\notin W}H^0(L_w,T)$, so $H^0(U,\mathscr{T})$ is discrete in $\bigoplus_{v\notin U}H^0(K_v,T)$ because it is the intersection of $H^0(W,\mathscr{T})\subset\bigoplus_{w\notin W}H^0(L_w,T)$ with $\bigoplus_{v\notin U}H^0(K_v,T)=(\bigoplus_{w\notin W}H^0(L_w,T))^G$. Thus I is discrete when $\mathscr{C}=\mathscr{T}$ is one single torus.

In the general case, exact triangle (11) yields a commutative diagram with exact lines

$$\begin{array}{ccccc}
H^1(U,\mathscr{M}) & \longrightarrow & H^0(U,\mathscr{C}) & \xrightarrow{u} & H^0(U,\mathscr{T}) \\
\downarrow & & \downarrow j & & \downarrow \\
\bigoplus_{v\notin U}H^1(K_v,M) & \longrightarrow & \bigoplus_{v\notin U}H^0(K_v,C) & \longrightarrow & \bigoplus_{v\notin U}H^0(K_v,T),
\end{array}$$

where $\mathscr{M}$ is a U-group of multiplicative type and $\mathscr{T}$ is a U-torus. Since $U\neq X$, the right vertical map is injective. As the lemma holds for $\mathscr{C}=\mathscr{T}$, the image J

of $H^0(U,\mathscr{T})$ into $\bigoplus_{v\notin U} H^0(K_v,T)$ is discrete, hence there is an open subgroup H of $\bigoplus_{v\notin U} H^0(K_v,T)$ such that $J\cap H=\{0\}$. Let H_1 be the inverse image of H in $\bigoplus_{v\notin U} H^0(K_v,C)$, it is an open subgroup of $\bigoplus_{v\notin U} H^0(K_v,C)$ such that $j^{-1}(H_1)$ is a subgroup of $\ker u$. As $H^1(U,\mathscr{M})$ is finite (Remark 3.2), we also have that $\ker u$ is finite and so is $j^{-1}(H_1)$. Therefore, $I\cap H_1=j(j^{-1}(H_1))$ is finite, which implies that I is discrete.

The same result for the image of $H^{-1}(U,\mathscr{C})$ into $\bigoplus_{v\notin U} H^{-1}(K_v,C)$ follows immediately because $H^{-1}(U,\mathscr{C})$ is a subgroup of $H^0(U,\mathscr{T}_1)$ (which has just been shown to be a discrete subgroup of $\bigoplus_{v\notin U} H^0(K_v,T_1)$), and $\bigoplus_{v\notin U} H^{-1}(K_v,C)$ is a topological subspace of $\bigoplus_{v\notin U} H^0(K_v,T_1)$. □

Remark 3.8. The analogue of Lemma 3.7 does not hold over a number field as soon as at least one finite place of K is not in U and K has at least two non archimedean places: indeed, for the exact sequence

$$0\to H^0(U,\mathbf{G}_m)\to\bigoplus_{v\notin U} H^0(K_v,\mathbf{G}_m)\to H^1_{\mathrm{c}}(U,\mathbf{G}_m)\to H^1(U,\mathbf{G}_m)\to 0$$

to hold (cf. [8, beginning of §2]), the groups $H^0(K_v,\mathbf{G}_m)$ at the archimedean places must be understood as the *modified* Tate group $\widehat{H}^0(K_v,\mathbf{G}_m)$. The intersection I of $H^0(U,\mathbf{G}_m)$ with the compact subgroup $\bigoplus_{v\notin U}\mathscr{O}_v^*\subset\bigoplus_{v\notin U} H^0(K_v,\mathbf{G}_m)$ (where by convention $\mathscr{O}_v^*$ means $\widehat{H}^0(K_v,\mathbf{G}_m)$ at the archimedean places) is countable and infinite (it is isomorphic to $\mathscr{O}_K^*$), hence I is not compact by Baire's Theorem. Therefore, the image of $H^0(U,\mathbf{G}_m)$ in $\bigoplus_{v\notin U} H^0(K_v,\mathbf{G}_m)$ is not closed.

Equip the finitely generated group $H^0(U,\mathscr{C})$ with the discrete topology. We give $H^0_{\mathrm{c}}(U,\mathscr{C})$ the unique topology such that all maps in the exact sequence

$$H^{-1}(U,\mathscr{C})\to\bigoplus_{v\notin U} H^{-1}(K_v,C)\to H^0_{\mathrm{c}}(U,\mathscr{C})\to H^0(U,\mathscr{C}) \tag{16}$$

are strict (by Lemma 3.7, the left map is strict and the quotient of $\bigoplus_{v\notin U} H^{-1}(K_v,C)$ by the image of $H^{-1}(U,\mathscr{C})$ is a locally compact Hausdorff group). We also give the finite group (cf. Proposition 3.6 (c) $D^1(U,\mathscr{C})$ the discrete topology, and topologize $H^1_{\mathrm{c}}(U,\mathscr{C})$ such that all maps in the exact sequence

$$H^0(U,\mathscr{C})\to\bigoplus_{v\notin U} H^0(K_v,C)\to H^1_{\mathrm{c}}(U,\mathscr{C})\to D^1(U,\mathscr{C})\to 0 \tag{17}$$

are strict.

Definition 3.9. Define $\mathscr{E}$ as the class of those abelian topological groups A that are an extension

$$0\to P\to A\xrightarrow{\pi} F\to 0 \tag{18}$$

(the maps being continuous) of a finitely generated group F (equipped with the discrete topology) by a profinite group P (this implies that all maps in this exact sequence are strict by [8, Lem. 3.4]).

It is easy to check that for every group A in $\mathscr{E}$, a closed subgroup of A and the quotient of A by any closed subgroup of A are still in $\mathscr{E}$. Also a topological extension of a (discrete) finitely generated group by A stays in $\mathscr{E}$. Finally, every group A in $\mathscr{E}$ is isomorphic to the direct product of a finitely generated group (equipped with discrete topology) by a profinite group: indeed, up to replacing F by F/F_{tors} and P by $\pi^{-1}(F_{\mathrm{tors}})$ in the extension (18), we can assume that $F=\mathbf{Z}^r$ for some $r\geq 0$. Since F is free, the morphism π has a section $s:F\to A$, which is automatically continuous because F is discrete. Setting $B=s(F)$, we get a topological isomorphism $A\cong P\times B$.

Definition 3.10. Let A be an abelian group. For every prime number ℓ, the *ℓ-adic completion* of A is

$$A^{(\ell)} := \varprojlim_{m\in\mathbf{N}} (A/\ell^m).$$

We also set

$$A_\wedge := \varprojlim_{n\in\mathbf{N}^*} A/n = \prod_{\ell \text{ prime}} A^{(\ell)}.$$

The piece of notation $A\{\ell\}$ stands for the ℓ-primary torsion subgroup of A.

For A finitely generated, we have $A^{(\ell)} = A\otimes_{\mathbf{Z}}\mathbf{Z}_\ell$; since $\mathbf{Z}_\ell$ is a torsion-free (hence flat) $\mathbf{Z}$-module, the functors $A\mapsto A^{(\ell)}$ and $A\mapsto A_\wedge$ are exact in the category of finitely generated abelian groups.

Lemma 3.11. *Let $A\to B\to E\to 0$ be an exact sequence of abelian groups.*

(a) *The induced map $B^{(\ell)}\to E^{(\ell)}$ is surjective.*
(b) *If ${}_\ell E$ is finite, then the induced sequence*

$$A^{(\ell)}\to B^{(\ell)}\to E^{(\ell)}\to 0$$

is exact. Likewise if A/ℓ is finite.

Proof. (a) Since the functor $.\otimes_{\mathbf{Z}}\mathbf{Z}/\ell^m$ is right exact, the sequence

$$A/\ell^m\to B/\ell^m\to E/\ell^m\to 0$$

is exact. Therefore, the projective system $(\ker[B/\ell^m\to E/\ell^m])_{m\geq 1}$ has surjective transition maps, which implies that the map $\varprojlim_m(B/\ell^m)\to\varprojlim_m(E/\ell^m)$ remains surjective.

(b) Assume that ${}_\ell E$ is finite. Then (by induction on m) we also have that ${}_{\ell^m}E$ is finite for every positive integer m thanks to the exact sequence

$${}_\ell E\to {}_{\ell^{m+1}}E\xrightarrow{\cdot\ell} {}_{\ell^m}E.$$

Let $I\subset B$ be the image of A by the map $A\to B$. By (a), the map $A^{(\ell)}\to I^{(\ell)}$ is surjective, so it is sufficient to prove that the sequence

$$I^{(\ell)}\to B^{(\ell)}\to E^{(\ell)}$$

is exact. By the snake lemma, we have an exact sequence

$${}_{\ell^m}E\to I/\ell^m\to B/\ell^m\to E/\ell^m.$$

Taking projective limit over m yields the required exact sequence because the kernel of the map $I/\ell^m\to B/\ell^m$ is finite (it is a quotient of ${}_{\ell^m}E$). Similarly, if A/ℓ is finite, then A/ℓ^m (hence also I/ℓ^m) is finite for every positive m and the same argument works. □

If we assume further that A is a topological abelian group, its *profinite completion* is $A^\wedge := \varprojlim_H(A/H)$, where H runs over all open subgroups of finite index in A. If A is profinite, then $A = A_\wedge = A^\wedge$. If A is in the class $\mathscr{E}$, then $A\hookrightarrow A_\wedge = A^\wedge$.

Proposition 3.12. *Let $\mathscr{C} = [\mathscr{T}_1\to\mathscr{T}_2]$ be a complex of U-tori with generic fiber $C = [T_1\to T_2]$. Let $v\in X^{(1)}$. The topological groups $H^{-1}(K_v, C)$ and $H^0(K_v, C)$ are in $\mathscr{E}$, as are the groups $H^0_{\mathrm{c}}(U,\mathscr{C})$ and $H^1_{\mathrm{c}}(U,\mathscr{C})$. In particular, for $i\in\{-1,0\}$, we have $H^i(K_v,C)_\wedge = H^i(K_v,C)^\wedge$ and for $i\in\{0,1\}$, we have $H^i_{\mathrm{c}}(U,\mathscr{C})\hookrightarrow H^i_{\mathrm{c}}(U,\mathscr{C})_\wedge$.*

Proof. Let $\mathscr{T}$ be a U-torus with generic fiber T. Let v be a closed point of X and let L be a finite Galois extension of K_v such that $\mathscr{T}$ splits over L. As $L^* \simeq \mathbf{Z} \times \mathscr{O}_L^*$ is in $\mathscr{E}$, so is $T(L)$. Then $H^0(K_v, T)$ is in $\mathscr{E}$ as a closed subgroup of $T(L)$ (the subgroup of $\mathrm{Gal}(L/K_v)$-invariants). The exact sequence

$$H^0(K_v, T_1) \to H^0(K_v, T_2) \to H^0(K_v, C) \to H^1(K_v, T_1)$$

and the definition of the topology on $H^0(K_v, C)$ now imply that $H^0(K_v, C)$ is in $\mathscr{E}$. The exact sequence (17) and Lemma 3.7 yield that $H^1_c(U, \mathscr{C})$ is in $\mathscr{E}$ because $D^1(U, \mathscr{C})$ is finite.

Similarly the group $H^{-1}(K_v, C) = \ker[H^0(K_v, T_1) \to H^0(K_v, T_2)]$ is a closed subgroup of $H^0(K_v, T_1)$, hence is in $\mathscr{E}$. This implies that $H^0_c(U, \mathscr{C})$ is in $\mathscr{E}$ thanks to the exact sequence (16), the group $H^0(U, \mathscr{C})$ being of finite type by Proposition 3.6. □

Lemma 3.13. *Let $v \in U$ and let $\mathscr{C}$ be a complex of $\mathscr{O}_v$-tori. Then $H^i(\mathscr{O}_v, \mathscr{C}) = 0$ for $i \geq 1$.*

Proof. Using the exact sequence (13) with U replaced by $\mathscr{O}_v$, we can assume that $\mathscr{C} = \mathscr{T}$ is one single torus. Since $\mathscr{T}$ is smooth over $\mathscr{O}_v$, the fppf cohomology group $H^i(\mathscr{O}_v, \mathscr{T})$ coincides with the étale group, and it is isomorphic ([16, Prop. II.1.1 (b)]) to the Galois cohomology group $H^i(\mathbf{F}_v, \tilde{T})$, where $\mathbf{F}_v$ is the residue field of $\mathscr{O}_v$ and $\tilde{T}$ the reduction of $\mathscr{T}$ mod. v, which is a torus over the finite field $\mathbf{F}_v$. Now $H^1(\mathbf{F}_v, \tilde{T}) = 0$ by Lang's theorem ([14]). For $i \geq 2$, the Galois cohomology group $H^i(\mathbf{F}_v, \tilde{T})$ is torsion. Let $n > 0$. By the Kummer sequence applied to the torus $\tilde{T}$ over the perfect field $\mathbf{F}_v$, the n-torsion subgroup ${}_nH^i(\mathbf{F}_v, \tilde{T})$ is a quotient of $H^i(\mathbf{F}_v, {}_nT)$, which is zero because $\mathbf{F}_v$ is of cohomological dimension 1 ([20, Chap. XIII, Prop. 2]). This proves the lemma. □

Remark 3.14. Using the definition of fppf compact support cohomology given in [8] (which, in particular, takes care of the set $\Omega_{\mathbf{R}}$ of real places; see loc. cit., Prop. 2.1), most results of this section hold (with the same proof) if we replace K by a number field with ring of integers $\mathscr{O}_K$, X by $\mathrm{Spec}\,\mathscr{O}_K$, and U by a non empty Zariski open subset of X. Also the piece of notation $v \notin U$ means that we consider the closed points of $X \setminus U$ *and* the real places of K; for $v \in \Omega_{\mathbf{R}}$ and $i \leq 0$, the groups $H^i(K_v, \dots)$ must be understood as the modified Tate groups (cf. Remark 3.8). More precisely:

- Lemma 3.1 (a) holds without the restriction $U \neq X$; (b) and Remark 3.2 are useless because for every finite flat commutative U-group scheme $\mathscr{N}$, all groups $H^i(U, \mathscr{N})$ and $H^i_c(U, \mathscr{N})$ are finite (cf. [8, Th. 1.1]).
- Lemma 3.4 (a) and (b) are unchanged; (c) holds without the restriction $U \neq X$. In (d), the vanishing of $H^i_c(U, \widehat{\mathscr{T}})$ for $i \geq 4$ still holds but the proof uses a different argument (namely that the dual of this group is the inverse limit of the $H^{4-i}(U, {}_n\mathscr{T})$, which is zero even in the case $i = 4$ because the finitely generated group $H^0(U, \mathscr{T})$ has trivial Tate module). The vanishing of $H^i(U, \widehat{\mathscr{T}})$ for $i \geq 4$ must be replaced by its finiteness if $\Omega_{\mathbf{R}} \neq \emptyset$. Finally, *the vanishing of $H^3(U, \widehat{\mathscr{T}})$ does not hold any more*, even if $\Omega_{\mathbf{R}} = \emptyset$ (if Leopoldt's conjecture is assumed, then $H^3(U, \widehat{\mathscr{T}})\{\ell\} = 0$ for ℓ invertible on U, but not in general for other primes; [16, Th. II.4.6 (b)] is wrong for $r = 3$, the problem in the proof being that the second line of the diagram needs not remain exact after taking profinite completions); neither does (e) hold as soon as there are at least two archimedean places. Also, there is no more counterexample as in Remark 3.5.

- Proposition 3.6 (a) and (c) hold (without the condition $U \neq X$), (b) is also true (and when $U = X$, the group $H^1(X, \mathscr{C})$ is even finite).

4. Duality theorems in fppf cohomology

In order to state and prove duality results for the cohomology of complexes of fppf sheaves, we need to extend some constructions from [8] to the context of bounded complexes. Let A and B be two bounded complexes of fppf sheaves over U. Following [21, XVII, 4.2.9] or [9, App. A], one can consider the Godement resolutions $G(A)$ and $G(B)$ of A and B. As in [12, II.6.6] or in [9, App. A], there is a natural commutative diagram of complexes

$$\begin{array}{ccc} A \otimes B & & \\ \downarrow & \searrow & \\ \mathrm{Tot}(G(A) \otimes G(B)) & \longrightarrow & G(A \otimes B). \end{array}$$

Following [8, Proof of Lem. 4.1], one gets a functorial morphism of complexes

$$\mathrm{Tot}(\Gamma_{\mathrm{c}}(U, G(A)) \otimes \Gamma(U, G(B))) \to \Gamma_{\mathrm{c}}(U, G(A \otimes B)) \tag{19}$$

and a functorial pairing

$$R\Gamma_{\mathrm{c}}(U, A) \otimes^{\mathbf{L}} R\Gamma(U, B) \to R\Gamma_{\mathrm{c}}(U, A \otimes B).$$

In particular, if C is a bounded complex and $\widehat{C} := \underline{\mathrm{Hom}}^{\bullet}(C, \mathbf{G}_m[1])$ its dual, using the morphism $\mathrm{Tot}(C \otimes \widehat{C}) \to \mathbf{G}_m[1]$ from Section 2, we get functorial pairings

$$\mathrm{Tot}(\Gamma_{\mathrm{c}}(U, G(C)) \otimes \Gamma(U, G(\widehat{C}))) \to \Gamma_{\mathrm{c}}(U, G(\mathbf{G}_m[1]))$$

and

$$R\Gamma_{\mathrm{c}}(U, C) \otimes^{\mathbf{L}} R\Gamma(U, \widehat{C}) \to R\Gamma_{\mathrm{c}}(U, \mathbf{G}_m[1]). \tag{20}$$

Following Section 2, given a bounded complex $\mathscr{C}$ of finite flat commutative group schemes over U, there is a natural topology on the abelian groups $H^i_{\mathrm{c}}(U, \mathscr{C})$. This topology is profinite via Lemma 2.3, and considering $H^j(U, \widehat{\mathscr{C}})$ as discrete torsion groups, the pairings

$$H^i_{\mathrm{c}}(U, \mathscr{C}) \times H^j(U, \widehat{\mathscr{C}}) \to H^{i+j+1}_{\mathrm{c}}(U, \mathbf{G}_m)$$

are continuous by the same argument as [8, Lem. 4.4].

Proposition 4.1. *Let $\mathscr{C}$ be a bounded complex of finite flat commutative group schemes over U. For all $i \in \mathbf{Z}$, there is a perfect pairing between profinite and discrete torsion groups*

$$H^{2-i}_{\mathrm{c}}(U, \mathscr{C}) \times H^i(U, \widehat{\mathscr{C}}) \to H^3_{\mathrm{c}}(U, \mathbf{G}_m) \cong \mathbf{Q}/\mathbf{Z}.$$

Proof. The isomorphism $H^3_{\mathrm{c}}(U, \mathbf{G}_m) \cong \mathbf{Q}/\mathbf{Z}$ follows from [16, Prop. II.2.6] and [8, Prop. 2.1 (4)].

We now prove the proposition by induction on the length of the complex $\mathscr{C}$.

- if $\mathscr{C}$ is concentrated in a given degree n, then the proposition is a direct consequence of [8, Th. 1.1].
- assume that $\mathscr{C} := [C_r \xrightarrow{f_r} C_{r+11} \to \cdots \to C_s]$, with C_i in degree i, has length $s - r \geq 1$. Let $A := \ker(f_r)$, and let $\overline{\mathscr{C}} := \mathrm{Cone}(A[-r] \xrightarrow{j} \mathscr{C})$. Then there is an exact triangle

$$A[-r] \xrightarrow{j} \mathscr{C} \xrightarrow{i} \overline{\mathscr{C}} \xrightarrow{p} A[1 - r].$$

We apply the functor $\widehat{\bullet}$ to this exact triangle. Then, using (1), we get that the natural triangle

$$\widehat{A[1-r]} \xrightarrow{\widehat{p}} \widehat{\overline{\mathscr{C}}} \xrightarrow{\widehat{i}} \widehat{\mathscr{C}} \xrightarrow{\widehat{j}} \widehat{A[-r]}$$

is exact.

Since the pairing between a complex and its dual is functorial, we get from the previous triangles a commutative diagram of topological groups with exact rows, where the vertical maps comes from the pairings (20):

$$\begin{array}{ccccc} H^{i-1}_{\mathrm{c}}(U,\overline{\mathscr{C}}) & \longrightarrow & H^{i-r}_{\mathrm{c}}(U,A) & \longrightarrow & H^{i}_{\mathrm{c}}(U,\mathscr{C}) \\ \downarrow & & \downarrow & & \downarrow \\ H^{3-i}(U,\widehat{\overline{\mathscr{C}}})^* & \longrightarrow & H^{3+r-i}(U,A^D)^* & \longrightarrow & H^{2-i}(U,\widehat{\mathscr{C}})^* \end{array} \tag{21}$$

$$\longrightarrow H^{i}_{\mathrm{c}}(U,\overline{\mathscr{C}}) \longrightarrow H^{i+1-r}_{\mathrm{c}}(U,A)$$
$$\downarrow \qquad\qquad \downarrow$$
$$\longrightarrow H^{2-i}(U,\widehat{\overline{\mathscr{C}}})^* \longrightarrow H^{2+r-i}(U,A^D)^*.$$

The second line remains exact as the dual of an exact sequence of discrete groups.

Note that we have a quasi-isomorphism $\varphi : \overline{\mathscr{C}} \to \mathscr{C}'$, where $\mathscr{C}' := [C_{r+1}/\operatorname{Im}(f_r) \to C_{r+2} \to \cdots \to C_s]$ has a smaller length than $\mathscr{C}$, hence by induction, we can assume that the natural maps $H^r_{\mathrm{c}}(U,\mathscr{C}') \to H^{2-r}(U,\widehat{\mathscr{C}'})^*$ are isomorphisms. Since all the C_i's are finite flat group schemes, we see that the dual morphism $\widehat{\varphi} : \widehat{\mathscr{C}'} \to \widehat{\overline{\mathscr{C}}}$ is a quasi-isomorphism, hence by functoriality of the pairings, we deduce that the maps $H^r_{\mathrm{c}}(U,\overline{\mathscr{C}}) \to H^{2-r}(U,\widehat{\overline{\mathscr{C}}})^*$ are isomorphisms too.

Hence in diagram (21), all vertical morphisms, except perhaps the central one, are isomorphisms. Then the five lemma implies that the central morphism is an isomorphism.

By induction on the length of $\mathscr{C}$, the proposition is proved. □

From now on, we denote by $\mathscr{C} = [\mathscr{T}_1 \to \mathscr{T}_2]$ a complex of U-tori with generic fiber $C = [T_1 \to T_2]$ and dual $\widehat{\mathscr{C}}$ (cf. Section 3). As a consequence of the previous proposition, we can get the global duality results for the cohomology of the complex $\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n$:

Proposition 4.2. *Let n be a positive integer (not necessarily prime to p). Let i be an integer with $-2 \leq i \leq 2$.*

(1) *There is a perfect pairing of finite groups*
$$H^i(U,\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n) \times H^{1-i}_{\mathrm{c}}(U,\widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n) \to \mathbf{Q}/\mathbf{Z}.$$
If $U \neq X$, the groups $H^2(U,\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n)$ and $H^{-1}_{\mathrm{c}}(U,\widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n)$ are zero.

(2) *There is a perfect pairing*
$$H^i_{\mathrm{c}}(U,\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n) \times H^{1-i}(U,\widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n) \to \mathbf{Q}/\mathbf{Z}$$
between the profinite group $H^i_{\mathrm{c}}(U,\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n)$ and the discrete group $H^{1-i}(U,\widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n)$. These groups are finite if $i \notin \{0,1\}$ or if p and n are coprime. The groups
$$H^{-2}_{\mathrm{c}}(U,\mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n) \quad \text{and} \quad H^3(U,\widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n)$$
are zero if $U \neq X$.

Moreover, all the groups involved are zero if $|i| > 2$.

Proof. Recall that there is a quasi-isomorphism of complexes $\psi = C' \to \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n$, where $C' := [{}_nT_1 \xrightarrow{\rho} {}_nT_2]$, with ${}_nT_1$ in degree -2, and that the dual morphism $\widehat{\psi} : \mathscr{C} \widehat{\otimes^{\mathbf{L}} \mathbf{Z}}/n \to \widehat{C'} = [\widehat{T_2}/n \xrightarrow{-\widehat{\rho}} \widehat{T_1}/n] = \big(\widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/n\big)[-1]$ (with $\widehat{T_2}/n$ in degree 0) is also a quasi-isomorphism.

Since C' is a bounded complex of finite flat commutative group schemes, then Proposition 4.1 implies that the pairings in the statement of the proposition are perfect pairings of topological groups.

Let us now check the finiteness and vanishing results. Using the exact triangle (9), we get an exact sequence:

$$H^{i+2}(U, {}_n\ker\rho) \to H^i(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n) \to H^{i+1}(U, T_{\mathbf{Z}/n}(\mathscr{C})).$$

As the finite U-group schemes $T_{\mathbf{Z}/n}(\mathscr{C})$ and ${}_n\ker\rho$ are of multiplicative type, Lemma 3.1 and Remark 3.2 imply that the groups $H^r(U, T_{\mathbf{Z}/n}(\mathscr{C}))$ and $H^r(U, {}_n\ker\rho)$ are finite for every integer r. Thus $H^i(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n)$ is finite.

Similarly, using the exact triangle (9), the group $H^i_{\mathrm{c}}(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n)$ is finite for $i \notin \{0, 1\}$ (or if p and n are coprime) by Remark 3.2.

Recall that for a finite and flat group scheme $\mathscr{N}$ over U, we have $H^r(U, \mathscr{N}) = 0$ for $r < 0$ (obvious), for $r \geq 4$, and also for $r = 3$ if $U \neq X$: indeed, by loc. cit., $H^r(U, \mathscr{N})$ is dual to $H^{3-r}_{\mathrm{c}}(U, \mathscr{N}^D)$; the latter is clearly zero if $r \geq 4$ and if $U \neq X$, we also have $H^0_{\mathrm{c}}(U, \mathscr{N}^D) = 0$ thanks to the exact sequence

$$0 \to H^0_{\mathrm{c}}(U, \mathscr{N}^D) \to H^0(U, \mathscr{N}^D) \to \bigoplus_{v \notin U} H^0(K_v, N^D),$$

the last map being injective by the assumption $U \neq X$. The previous dévissages now yield the vanishing assertions of the proposition. □

Proposition 4.3. *Let i be an integer. Let n be a positive integer (not necessarily prime to p).*

(a) *The groups ${}_nH^i(U, \mathscr{C})$ and $H^i(U, \mathscr{C})/n$ are finite. The group $H^i(U, \mathscr{C})$ is torsion if $i \geq 2$ (resp. if $i \geq 1$ and $U \neq X$). Besides $H^i(U, \mathscr{C}) = 0$ in the following cases: $i \geq 4$; $i \leq -2$; $i = 3$ and $U \neq X$.*

(b) *The groups ${}_nH^i_{\mathrm{c}}(U, \widehat{\mathscr{C}})$ and $H^i_{\mathrm{c}}(U, \widehat{\mathscr{C}})/n$ are finite. The group $H^i_{\mathrm{c}}(U, \widehat{\mathscr{C}})$ is torsion if $i \geq 2$, and it is zero if $i \geq 4$ or $i \leq -2$. Assume further $U \neq X$; then $H^i_{\mathrm{c}}(U, \widehat{\mathscr{C}}) = 0$ for $i = -1$.*

Proof. (a) Using the exact triangle in $\mathscr{D}^b(U)$:

$$\mathscr{C} \xrightarrow{\cdot n} \mathscr{C} \to \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n \to \mathscr{C}[1],$$

we get an exact sequence of abelian groups

$$0 \to H^i(U, \mathscr{C})/n \to H^i(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n) \to {}_nH^{i+1}(U, , \mathscr{C}) \to 0. \tag{22}$$

since $H^i(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n)$ is finite (Proposition 4.2 (1)), the finiteness of the groups ${}_nH^{i+1}(U, \mathscr{C})$ and $H^i(U, \mathscr{C})/n$ follows for all i.

To prove that $H^i(U, \mathscr{C})$ is torsion if $i \geq 2$ (resp. if $U \neq X$ and $i = 1$), we can restrict by dévissage (using exact sequence (13)) to the case when $\mathscr{C} = \mathscr{T}$ is one single torus. If $U \neq X$ and $i = 1$, this follows from Lemma 3.1 (a), so assume $i \geq 2$. We can also assume by a restriction-corestriction argument that $\mathscr{T} = \mathbf{G}_m$ because the torus $\mathscr{T}$ is split by some finite étale covering of U. Now $H^2(U, \mathbf{G}_m) = \mathrm{Br}\, U$ is torsion because it injects into $\mathrm{Br}\, K$; also $H^3(U, \mathbf{G}_m)$ is torsion (it is even 0 if $U \neq X$) and $H^i(U, \mathbf{G}_m) = 0$ for $i \geq 4$ by [16, Prop. II.2.1], the group scheme $\mathbf{G}_m$ being smooth (hence étale and fppf cohomology coincide).

For every U-torus $\mathscr{T}$, we have $H^i(U,\mathscr{T})=0$ for negative i (obvious), hence by dévissage $H^i(U,\mathscr{C})=0$ for $i<-1$. Let $i\geq 3$; as seen before $H^i(U,\mathscr{C})$ is torsion and ${}_nH^i(U,\mathscr{C})$ is a quotient of $H^{i-1}(U,\mathscr{C}\otimes^{\mathbf{L}}\mathbf{Z}/n)$ by exact sequence (22). The latter is zero if $i\geq 4$, and also if $i=3$ when $U\neq X$ by the vanishing assertions in Proposition 4.2 (1). Thus $H^i(U,\mathscr{C})$ is zero if $i\geq 4$, and also if $i=3$ if we assume further $U\neq X$.

(b) Similarly, the finiteness statements follow from the exact sequence

$$0\to H^i_{\mathrm{c}}(U,\widehat{\mathscr{C}})/n\to H^i_{\mathrm{c}}(U,\widehat{\mathscr{C}}\otimes^{\mathbf{L}}\mathbf{Z}/n)\to {}_nH^{i+1}_{\mathrm{c}}(U,\widehat{\mathscr{C}})\to 0 \tag{23}$$

combined with Proposition 4.2 (1). Let $i\geq 2$. To prove that $H^i_{\mathrm{c}}(U,\widehat{\mathscr{C}})$ is torsion we can assume that $\widehat{\mathscr{C}}$ is the dual of a torus (via exact sequence (14)), then that $\widehat{\mathscr{C}}=\mathbf{Z}$ (by a restriction-corestriction argument). Using the exact sequence

$$\bigoplus_{v\notin U}H^{i-1}(K_v,\mathbf{Z})\to H^i_{\mathrm{c}}(U,\mathbf{Z})\to H^i(U,\mathbf{Z}),$$

it is sufficient to prove that $H^i(U,\mathbf{Z})$ is torsion because the Galois cohomology groups $H^{i-1}(K_v,\mathbf{Z})$ are torsion for $i-1>0$. This holds by [16, Lem. II.2.10].

Let $\mathscr{T}$ be a U-torus. For each integer i, there is an exact sequence

$$\bigoplus_{v\notin U}H^{i-1}(K_v,\widehat{T})\to H^i_{\mathrm{c}}(U,\widehat{\mathscr{T}})\to H^i(U,\widehat{\mathscr{T}})\to\bigoplus_{v\notin U}H^i(K_v,\widehat{T}). \tag{24}$$

Therefore, $H^i_{\mathrm{c}}(U,\widehat{\mathscr{T}})=0$ for $i<0$, hence $H^i_{\mathrm{c}}(U,\widehat{\mathscr{C}})=0$ (by dévissage) for $i<-1$. For $i\geq 4$, we have $H^i_{\mathrm{c}}(U,\widehat{\mathscr{T}})=0$ by Lemma 3.4 (d), and $H^i_{\mathrm{c}}(U,\widehat{\mathscr{C}})=0$ by dévissage.

Assume now $U\neq X$. Then $H^0_{\mathrm{c}}(U,\widehat{\mathscr{T}})=0$ by exact sequence (24) applied to $i=0$: indeed, the map $H^0(U,\widehat{\mathscr{T}})\to\bigoplus_{v\notin U}H^0(K_v,\widehat{T})$ is injective (choose a closed point v of $X\setminus U$; then the restriction maps $H^0(U,\widehat{\mathscr{T}})\to H^0(K,\widehat{T})$ and $H^0(K,\widehat{T})\to H^0(K_v,\widehat{T})$ are injective). Therefore, $H^{-1}_{\mathrm{c}}(U,\widehat{\mathscr{C}})=0$ by dévissage. □

Proposition 4.4. *Let i be an integer. Let n be a positive integer (not necessarily prime to p).*

(a) *The group ${}_nH^i(U,\widehat{\mathscr{C}})$ is finite if $i\notin\{1,2\}$. The group $H^i(U,\widehat{\mathscr{C}})/n$ is finite if $i\neq 1$. The group $H^i(U,\widehat{\mathscr{C}})$ is torsion if $i\geq 1$, and it is zero if $i\geq 4$ or $i\leq -2$. If we assume further $U\neq X$, then $H^3(U,\widehat{\mathscr{C}})=0$.*

(b) *The group ${}_nH^i_{\mathrm{c}}(U,\mathscr{C})$ is finite if $i\neq 1$. The group $H^i_{\mathrm{c}}(U,\mathscr{C})/n$ is finite if $i\notin\{0,1\}$. The group $H^i_{\mathrm{c}}(U,\mathscr{C})$ is torsion for $i\geq 2$, and it is zero if $i\geq 4$ or $i\leq -2$ (resp. $i=-1$ if $U\neq X$).*

Proof. (a) The exact sequence

$$0\to H^i(U,\widehat{\mathscr{C}})/n\to H^i(U,\widehat{\mathscr{C}}\otimes^{\mathbf{L}}\mathbf{Z}/n)\to {}_nH^{i+1}(U,\widehat{\mathscr{C}})\to 0 \tag{25}$$

and Proposition 4.2 (2) yield the finiteness of ${}_nH^i(U,\widehat{\mathscr{C}})$ for $i\notin\{1,2\}$ and of $H^i(U,\widehat{\mathscr{C}})/n$ for $i\notin\{0,1\}$. Besides the abelian group $H^0(U,\widehat{\mathscr{C}})$ is of finite type by Proposition 3.6. In particular $H^0(U,\widehat{\mathscr{C}})/n$ is finite.

Let $i\geq 1$. To prove that $H^i(U,\widehat{\mathscr{C}})$ is torsion, we can assume (by dévissage) that $\widehat{\mathscr{C}}=\widehat{\mathscr{T}}$, where $\mathscr{T}$ is one single torus, then that $\widehat{\mathscr{C}}=\mathbf{Z}$ (by restriction-corestriction); then the result holds by [16, Lem. II.2.10]. For

$i \leq -2$ or $i \geq 4$ (resp. $i = 3$ if $U \neq X$), the group $H^i(U, \widehat{\mathscr{C}})$ is zero by dévissage (using Lemma 3.4 (d) for the latter).

(b) There is an exact sequence

$$0 \to H^i_{\mathrm{c}}(U, \mathscr{C})/n \to H^i_{\mathrm{c}}(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n) \to {}_nH^{i+1}_{\mathrm{c}}(U, \mathscr{C}) \to 0 \tag{26}$$

By Proposition 4.2 (2), the group ${}_nH^i_{\mathrm{c}}(U, \mathscr{C})$ is finite for $i \notin \{1, 2\}$ and the group $H^i_{\mathrm{c}}(U, \mathscr{C})/n$ is finite for $i \notin \{0, 1\}$. To prove that the groups $H^i_{\mathrm{c}}(U, \mathscr{C})$ are torsion for $i \geq 2$, we can assume as usual that $\mathscr{C} = \mathscr{T}$ is a torus. Then we apply Proposition 4.3 (a) and the exact sequence

$$\bigoplus_{v \notin U} H^{i-1}(K_v, T) \to H^i_{\mathrm{c}}(U, \mathscr{T}) \to H^i(U, \mathscr{T}).$$

Besides $H^2_{\mathrm{c}}(U, \mathscr{T})$ is finite by Lemma 3.4 (b), so $H^2_{\mathrm{c}}(U, \mathscr{C})$ is also of cofinite type by dévissage because we already know that $H^3_{\mathrm{c}}(U, \mathscr{T}_1)$ is torsion of cofinite type.

Obviously we have $H^i_{\mathrm{c}}(U, \mathscr{T}) = 0$ for every negative i, hence $H^i_{\mathrm{c}}(U, \mathscr{C}) = 0$ by dévissage if $i \leq -2$. Assume $i \geq 4$. then $H^i_{\mathrm{c}}(U, \mathscr{C}) = H^i(U, \mathscr{C})$ (apply the exact sequence (2) and use the fact that K_v is of strict cohomological dimension 2), so $H^i_{\mathrm{c}}(U, \mathscr{C}) = 0$ by Proposition 4.3 (a). If we assume further $U \neq X$, then $H^0_{\mathrm{c}}(U, \mathscr{T}) = 0$ (same argument as in Proposition 4.3 (b), so $H^{-1}_{\mathrm{c}}(U, \mathscr{C}) = 0$ by dévissage. □

Remark 4.5. Using Remark 3.2, it is easy to see that the finiteness assertions of Proposition 4.4 hold for every i if we assume further that p does not divide n, but this is no longer true in general if $U \neq X$. Indeed the group $H^1(U, \mathbf{Z}/p)$ and its dual $H^2_{\mathrm{c}}(U, \mu_p)$ can be infinite (cf. [16, Lem. III.8.9]). Since $H^1(U, \mathbf{Z}) = 0$ and ${}_pH^2_{\mathrm{c}}(U, \mathbf{G}_m)$ is finite, this implies that ${}_pH^2(U, \mathbf{Z})$ and $H^1_{\mathrm{c}}(U, \mathbf{G}_m)/p$ are infinite, which gives examples of ${}_pH^i(U, \widehat{\mathscr{C}})$ infinite for $i = 1, 2$ and of $H^i_{\mathrm{c}}(U, \mathscr{C})/p$ infinite for $i = 0, 1$. The complex $\mathscr{C} = [\mathbf{G}_m \xrightarrow{\cdot p} \mathbf{G}_m]$ is an example with ${}_pH^1_{\mathrm{c}}(U, \mathscr{C})$ and $H^1(U, \widehat{\mathscr{C}})/p$ infinite (indeed, $\mathscr{C}$ is quasi-isomorphic to $\mu_p[1]$ and $\widehat{\mathscr{C}}$ is quasi-isomorphic to $\mathbf{Z}/p$).

Remark 4.6. For every integer r and every positive integer n, the groups $H^r(U, \mathscr{C})/n$ and $H^r_{\mathrm{c}}(U, \widehat{\mathscr{C}})/n$ are finite by Proposition 4.3 (b), so for each prime number ℓ (including $\ell = p$), the ℓ-adic completions

$$H^r(U, \mathscr{C})^{(\ell)} := \varprojlim_m H^r(U, \mathscr{C})/\ell^m; \quad H^r_{\mathrm{c}}(U, \widehat{\mathscr{C}})^{(\ell)} := \varprojlim_m H^r_{\mathrm{c}}(U, \widehat{\mathscr{C}})/\ell^m$$

are profinite. Exact sequence (26) shows that $H^r_{\mathrm{c}}(U, \mathscr{C})/n$ is a closed subgroup of the profinite group $H^r_{\mathrm{c}}(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/n)$, hence $H^r_{\mathrm{c}}(U, \mathscr{C})/n$ is profinite and so is the ℓ-adic completion $H^r_{\mathrm{c}}(U, \mathscr{C})^{(\ell)}$.

The map $\mathscr{C} \otimes^{\mathbf{L}} \widehat{\mathscr{C}} \to \mathbf{G}_m[1]$ induces for every integer r pairings

$$H^r(U, \mathscr{C}) \times H^{2-r}_{\mathrm{c}}(U, \widehat{\mathscr{C}}) \to H^3_{\mathrm{c}}(U, \mathbf{G}_m) \cong \mathbf{Q}/\mathbf{Z}. \tag{27}$$

$$H^r_{\mathrm{c}}(U, \mathscr{C}) \times H^{2-r}(U, \widehat{\mathscr{C}}) \to H^3_{\mathrm{c}}(U, \mathbf{G}_m) \cong \mathbf{Q}/\mathbf{Z}. \tag{28}$$

We now prove a key lemma.

Lemma 4.7. *Let ℓ be a prime number (possibly equal to p). Let i be an integer.*

(a) *The maps*

$$\psi : H^{i+1}(U, \mathscr{C})\{\ell\} \to (H^{1-i}_{\mathrm{c}}(U, \widehat{\mathscr{C}})^{(\ell)})^*$$
$$\psi' : H^{1-i}_{\mathrm{c}}(U, \widehat{\mathscr{C}})\{\ell\} \to (H^{i+1}(U, \mathscr{C})^{(\ell)})^*$$

induced by the pairing (27) *are surjective and have divisible kernel. Besides* ψ' *is an isomorphism if* $i = -2$ *and* ψ *is an isomorphism if we have both* $i = 1$ *and* $U \neq X$.

(b) *The map*

$$\varphi : H^{i+1}(U, \widehat{\mathscr{C}})\{\ell\} \to (H_{\mathrm{c}}^{1-i}(U, \mathscr{C})^{(\ell)})^*$$

induced by the pairing (28) *is surjective, and has divisible kernel (resp. is an isomorphism if we assume both* $i = 1$ *and* $U = X$*). Assume* $i \notin \{-1, 0\}$. *Then the map*

$$\varphi' : H_{\mathrm{c}}^{1-i}(U, \mathscr{C})\{\ell\} \to (H^{i+1}(U, \widehat{\mathscr{C}})^{(\ell)})^*$$

is surjective and has divisible kernel (resp. is an isomorphism if $i = -2$*).*

Observe that for $U \neq X$, the groups involved can be non zero only if $-2 \leq i \leq 1$.

Proof. (a) For each positive integer m, there is an exact commutative diagram of finite abelian groups,

(29)

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^i(U, \mathscr{C})/\ell^m & \longrightarrow & H^i(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/\ell^m) & \longrightarrow & {}_{\ell^m} H^{i+1}(U, \mathscr{C}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \psi_m \\ 0 & \longrightarrow & ({}_{\ell^m} H_{\mathrm{c}}^{2-i}(U, \widehat{\mathscr{C}}))^* & \longrightarrow & H_{\mathrm{c}}^{1-i}(U, \widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/\ell^m)^* & \longrightarrow & (H_{\mathrm{c}}^{1-i}(U, \widehat{\mathscr{C}})/\ell^m)^* \longrightarrow 0. \end{array}$$

By Proposition 4.2 (1), the middle vertical map is an isomorphism. Taking direct limit over m and applying the snake lemma, we get that $\psi = \varinjlim_m \psi_m$ is surjective and $\operatorname{Ker} \psi$ is a quotient of $(T_\ell H_{\mathrm{c}}^{2-i}(U, \widehat{\mathscr{C}}))^*$. Since each ${}_{\ell^m} H_{\mathrm{c}}^{2-i}(U, \widehat{\mathscr{C}})$ is finite, the ℓ-adic Tate module $T_\ell H_{\mathrm{c}}^{2-i}(U, \widehat{\mathscr{C}})$ is profinite and torsion-free, which implies that the dual $(T_\ell H_{\mathrm{c}}^{2-i}(U, \widehat{\mathscr{C}}))^*$ is divisible. For $U \neq X$ and $i = 1$, the group $H_{\mathrm{c}}^1(U, \widehat{\mathscr{C}})$ is of finite type by Proposition 3.6 (a), so its ℓ-adic Tate-module is zero and ψ has trivial kernel.

The argument for ψ' is similar, using the exact commutative diagram

(30)

$$\begin{array}{ccccccc} 0 & \longrightarrow & H_{\mathrm{c}}^{-i}(U, \widehat{\mathscr{C}})/\ell^m & \longrightarrow & H_{\mathrm{c}}^{-i}(U, \widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/\ell^m) & \longrightarrow & {}_{\ell^m} H_{\mathrm{c}}^{1-i}(U, \widehat{\mathscr{C}}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \psi'_m \\ 0 & \longrightarrow & ({}_{\ell^m} H^{2+i}(U, \mathscr{C}))^* & \longrightarrow & H^{2+i}(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/\ell^m)^* & \longrightarrow & (H^{1+i}(U, \mathscr{C})/\ell^m)^* \longrightarrow 0. \end{array}$$

Besides, for $i = -2$, the Tate module of $H^{2+i}(U, C) = H^0(U, \mathscr{C})$ is trivial because $H^0(U, \mathscr{C})$ is a finitely generated abelian group (Proposition 3.6 (b)), which gives that ψ' has trivial kernel.

(b) There is an exact commutative diagram of discrete abelian groups (observe that the second line is obtained by dualizing an exact sequence of profinite groups):

(31)

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^i(U, \widehat{\mathscr{C}})/\ell^m & \longrightarrow & H^i(U, \widehat{\mathscr{C}} \otimes^{\mathbf{L}} \mathbf{Z}/\ell^m) & \longrightarrow & {}_{\ell^m} H^{i+1}(U, \widehat{\mathscr{C}}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \varphi_m \\ 0 & \longrightarrow & ({}_{\ell^m} H_{\mathrm{c}}^{2-i}(U, \mathscr{C}))^* & \longrightarrow & H_{\mathrm{c}}^{1-i}(U, \mathscr{C} \otimes^{\mathbf{L}} \mathbf{Z}/\ell^m)^* & \longrightarrow & (H_{\mathrm{c}}^{1-i}(U, \mathscr{C})/\ell^m)^* \longrightarrow 0. \end{array}$$

Since the middle vertical is an isomorphism by Proposition 4.2 (2) and ${}_{\ell^m}(H_{\mathrm{c}}^{2-i}(U, \mathscr{C}))$ is profinite for each m (hence $H_{\mathrm{c}}^{2-i}(U, \mathscr{C})$ has profinite ℓ-adic Tate module), the same argument as in (a) yields that φ is surjective

with divisible kernel. If $U = X$ and $i = 1$, then $H^{2-i}_c(U, \mathscr{C}) = H^1(X, \mathscr{C})$ is of finite type by Proposition 3.6 (b), so it has trivial ℓ-adic Tate module and φ is an isomorphism.

The argument for φ' is similar, except that we use the exact commutative diagram

(32)
$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^{-i}_c(U,\mathscr{C})/\ell^m & \longrightarrow & H^{-i}_c(U,\mathscr{C}\otimes^{\mathbf{L}}\mathbf{Z}/\ell^m) & \longrightarrow & {}_{\ell^m}H^{1-i}_c(U,\mathscr{C}) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow{\scriptstyle \varphi'_m} & & \\ 0 & \longrightarrow & ({}_{\ell^m}H^{i+2}(U,\widehat{\mathscr{C}}))^* & \longrightarrow & H^{i+1}(U,\widehat{\mathscr{C}}\otimes^{\mathbf{L}}\mathbf{Z}/\ell^m)^* & \longrightarrow & (H^{i+1}(U,\widehat{\mathscr{C}})/\ell^m)^* & \longrightarrow & 0. \end{array}$$

only for $i \notin \{-1, 0\}$ (for $i \in \{-1, 0\}$ and $U \neq X$, the diagram would consist of profinite but possibly infinite groups if $\ell = p$, so direct limits would not necessarily behave well; in particular, ℓ-adic completions involved would not necessarily be profinite). The same argument as in (a) shows that φ' is surjective with divisible kernel, and this kernel is trivial for $i = -2$ because the finitely generated abelian group $H^0(U, \widehat{\mathscr{C}})$ (cf. Proposition 3.6 (a)) has trivial ℓ-adic Tate module. □

Remark 4.8. For abelian groups A, B, assertions like "$A\{\ell\} \to (B^{(\ell)})^*$ is surjective with divisible kernel" can be rephrased as follows: the pairing $A\{\ell\} \times B^{(\ell)} \to \mathbf{Q}/\mathbf{Z}$ has trivial right kernel and divisible left kernel.

The following theorem extends the function field case of [16, Th. II.4.6]. (which corresponds to $\mathscr{C} = \mathscr{T}$ or $\mathscr{C} = \mathscr{T}[1]$, where $\mathscr{T}$ is a torus).

Theorem 4.9. (a) *The pairing* (27) *induces a perfect duality between the discrete torsion group $H^3_c(U, \widehat{\mathscr{C}})$ and the finite-type $\widehat{\mathbf{Z}}$-module $H^{-1}(U, \mathscr{C})_\wedge$, resp. between the discrete torsion group $H^2_c(U, \widehat{\mathscr{C}})$ and the finite-type $\widehat{\mathbf{Z}}$-module $H^0(U, \mathscr{C})_\wedge$.*

(b) *Assume $U \neq X$. The pairing* (27) *induces a perfect duality between the discrete torsion group $H^1(U, \mathscr{C})$ and the finite-type $\widehat{\mathbf{Z}}$-module $H^1_c(U, \widehat{\mathscr{C}})_\wedge$, resp. between the discrete torsion group $H^2(U, \mathscr{C})$ and the finite-type $\widehat{\mathbf{Z}}$-module $H^0_c(U, \widehat{\mathscr{C}})_\wedge$.*

Proof. (a) Let ℓ be a prime number. Then the map ψ' of Lemma 4.7 (a) is an isomorphism for $i = -2$, which yields the first point (recall that $H^3_c(U, \widehat{\mathscr{C}})$ and $H^2_c(U, \widehat{\mathscr{C}})$ are torsion of cofinite type by Proposition 4.3; also $H^{-1}(U, \mathscr{C})$ and $H^0(U, \mathscr{C})$ are finitely generated by Proposition 3.6).

In the case $U = X$, the second point is a duality between $H^2(X, \widehat{\mathscr{C}})$ and $H^0(X, C)_\wedge$, which follows from Lemma 4.7 (b) in the case $i = 1$. Now assume $U \neq X$ and let $\mathscr{T}$ be a U-torus. By the first point applied to $\mathscr{C} = \mathscr{T}[1]$, the group $H^3_c(U, \widehat{\mathscr{T}})$ is dual to $H^0(U, \mathscr{T})_\wedge$. By Lemma 4.7 (a) with $i = -1$ and $\mathscr{C} = \mathscr{T}[1]$, the finite group $H^2_c(U, \widehat{\mathscr{T}})$ (cf. Lemma 3.4 (c) is dual to the finite group $H^1(U, T)$ (cf. Lemma 3.1 (a): indeed, a finite group coincides with its ℓ-adic completion and doesn't contain a non trivial divisible subgroup. There is a commutative diagram with exact lines (observe that the second line is obtained by applying the ℓ-completion functor to an exact sequence of finitely generated group, then dualizing an exact

sequence of profinite groups):

$$\begin{array}{ccccccccc} H^2_c(U,\widehat{\mathscr{T}_2})\{\ell\} & \longrightarrow & H^2_c(U,\widehat{\mathscr{T}_2})\{\ell\} & \longrightarrow & H^2_c(U,\widehat{\mathscr{C}})\{\ell\} & \longrightarrow & H^3_c(U,\widehat{\mathscr{T}_2})\{\ell\} & \longrightarrow & H^3_c(U,\widehat{\mathscr{T}_1})\{\ell\} \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle g_1} & & \downarrow{\scriptstyle g_2} \\ (H^1(U,\mathscr{T}_2)^{(\ell)})^* & \to & (H^1(U,\mathscr{T}_1)^{(\ell)})^* & \to & (H^0(U,\mathscr{C})^{(\ell)})^* & \to & (H^0(U,\mathscr{T}_2)^{(\ell)})^* & \to & (H^0(U,\mathscr{T}_1)^{(\ell)})^*. \end{array}$$

Now h is an isomorphism by the five-lemma, whence the result.

(b) Consider diagram (29) for $i = 0$. By (a), the left vertical map is an isomorphism and by Proposition 4.2 (1), the middle vertical map is an isomorphism, hence ψ_m is an isomorphism from ${}_{\ell^m}H^1(U,\mathscr{C})$ to $(H^1_c(U,\widehat{\mathscr{C}})/\ell^m)^*$. Taking direct limit over m, then direct sum over all prime ℓ, yields the duality between the torsion group of cofinite type (cf. Proposition 4.3) $H^1(U,\mathscr{C})$ and the finite type $\widehat{\mathbf{Z}}$-module (cf. Proposition 3.6) $H^1_c(U,\widehat{\mathscr{C}})_\wedge$.

Lemma 4.7 (a) for $i = 1$ yields that for $U \neq X$, the map ψ an isomorphism, which immediately gives the duality between the torsion group of finite type (cf. Proposition 4.3) $H^2(U,\mathscr{C})$ and the finite type $\widehat{\mathbf{Z}}$-module (cf. Proposition 3.6) $H^0_c(U,\widehat{\mathscr{C}})_\wedge$. □

Remark 4.10. In the case $U = X$, the first assertion of Theorem 4.9 (b) should be replaced by a duality between $H^1(X,\widehat{\mathscr{C}})$ and $H^1(X,\mathscr{C})_\wedge$ (see Theorem 4.11 (b) below in the case $U = X$). The second assertion (duality between $H^2(X,\mathscr{C})$ and $H^0(X,\widehat{\mathscr{C}})_\wedge$) actually still holds, cf. Theorem 4.11 (a).

Theorem 4.11. (a) *The pairing* (28) *induces a perfect duality between the discrete torsion group $H^3_c(U,\mathscr{C})$ and the finite-type $\widehat{\mathbf{Z}}$-module $H^{-1}(U,\widehat{\mathscr{C}})_\wedge$, resp. between the discrete torsion group $H^2_c(U,\mathscr{C})$ and the finite-type $\widehat{\mathbf{Z}}$-module $H^0(U,\widehat{\mathscr{C}})_\wedge$.*

(b) *The pairing* (28) *induces a perfect duality between the discrete torsion group $H^1(U,\widehat{\mathscr{C}})$ and the profinite group $H^1_c(U,\mathscr{C})_\wedge$, resp. between the discrete torsion group $H^2(U,\widehat{\mathscr{C}})$ and the profinite group $H^0_c(U,\mathscr{C})_\wedge$.*

Proof. (a) Let ℓ be any prime number. The map φ' of Lemma 4.7 (b) is an isomorphism for $i = -2$, which yields the first point (Proposition 4.2 (2) yields that $H^3_c(U,\mathscr{C})$ and $H^2_c(U,\mathscr{C})$ are torsion groups of cofinite type; Proposition 3.6 gives that $H^{-1}(U,\widehat{\mathscr{C}})$ and $H^0(U,\widehat{\mathscr{C}})$ are finitely generated).

There is a commutative diagram with exact lines

$$\begin{array}{ccccccccc} H^2_c(U,\mathscr{T}_1)\{\ell\} & \longrightarrow & H^2_c(U,\mathscr{T}_2)\{\ell\} & \longrightarrow & H^2_c(U,\mathscr{C})\{\ell\} & \longrightarrow & H^3_c(U,\mathscr{T}_1)\{\ell\} & \longrightarrow & H^3_c(U,\mathscr{T}_2)\{\ell\} \\ \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} & & \downarrow{\scriptstyle h} & & \downarrow{\scriptstyle g_1} & & \downarrow{\scriptstyle g_2} \\ (H^1(U,\widehat{\mathscr{T}_1})^{(\ell)})^* & \to & (H^1(U,\widehat{\mathscr{T}_2})^{(\ell)})^* & \to & (H^0(U,\widehat{\mathscr{C}})^{(\ell)})^* & \to & (H^0(U,\widehat{\mathscr{T}_1})^{(\ell)})^* & \to & (H^0(U,\widehat{\mathscr{T}_2})^{(\ell)})^*. \end{array}$$

The maps g_1 and g_2 are isomorphisms by the first point applied to $\mathscr{C} = \mathscr{T}_1$, $\mathscr{C} = \mathscr{T}_2$. The maps f_1 and f_2 are isomorphisms by Lemma 4.7 (b) applied to the same complexes (map φ in the case $i = -1$): indeed, for a U-torus $\mathscr{T}$, the groups $H^1(U,\widehat{\mathscr{T}})$ and $H^2_c(U,\mathscr{T})$ are finite (Lemma 3.4 (a) and (b), hence they coincide with their ℓ-adic completions and do not contain a non trivial divisible subgroup. Therefore, h is an isomorphism by the five-lemma, whence the second point.

(b) Consider diagram (31) for $i = 0$. By (a), the left vertical map is an isomorphism and the middle vertical map is an isomorphism by Proposition 4.2 (1), hence φ_m is an isomorphism from ${}_{\ell^m}H^1(U,\widehat{\mathscr{C}})$ to $(H^1_c(U,\mathscr{C})/\ell^m)^*$. Taking

direct limit over m, then direct sum over all prime ℓ, yields the duality between $H^1(U,\widehat{\mathscr{C}})$ (which is torsion by Proposition 4.4, but not necessarily of cofinite type, cf. Remark 4.5) and $H^1_{\mathrm{c}}(U,\mathscr{C})_\wedge$.

Now consider diagram (31) for $i=1$. By the previous duality, the left vertical map induces an isomorphism between $H^1(U,\widehat{\mathscr{C}})/\ell^m$ and $(_{\ell^m}(H^1_{\mathrm{c}}(U,\mathscr{C})_\wedge))^*$. Since $H^1_{\mathrm{c}}(U,\mathscr{C})$ is in the class $\mathscr{E}$ (that is: it is the product of a finite type group by a profinite group) by Proposition 3.12, the ℓ^m-torsion of $H^1_{\mathrm{c}}(U,\mathscr{C})$ and of $H^1_{\mathrm{c}}(U,\mathscr{C})_\wedge$ coincide, hence the left vertical map is actually an isomorphism and the right vertical map ψ_m is an isomorphism as well (the middle vertical map is an isomorphism by Proposition 4.2 (2)). Taking direct limit and direct sum over all prime ℓ, we get the duality between the torsion group (cf. Proposition 4.4) $H^2(U,\widehat{\mathscr{C}})$ and the profinite group $H^0_{\mathrm{c}}(U,\mathscr{C})_\wedge$. □

Proposition 4.12. *The pairing* (27) *for* $r=1$ *induces a perfect pairing of finite groups*

$$D^1(U,\mathscr{C})\times D^1(U,\widehat{\mathscr{C}})\to \mathbf{Q}/\mathbf{Z}. \tag{33}$$

Proof. Fix a prime number ℓ. There is a commutative diagram:

(34)

$$\begin{array}{ccccccc} 0 & \longrightarrow & D^1(U,\mathscr{C})\{\ell\} & \longrightarrow & H^1(U,\mathscr{C})\{\ell\} & \longrightarrow & \bigoplus_{v\notin U} H^1(K_v,C)\{\ell\} \\ & & & & \downarrow\psi & & \downarrow\beta \\ 0 & \longrightarrow & (D^1(U,\widehat{\mathscr{C}})^{(\ell)})^* & \longrightarrow & (H^1_{\mathrm{c}}(U,\widehat{\mathscr{C}})^{(\ell)})^* & \longrightarrow & (\bigoplus_{v\notin U} H^0(K_v,\widehat{C})^{(\ell)})^*. \end{array}$$

The first line is exact by definition of $D^1(U,\mathscr{C})$. The sequence

$$\bigoplus_{v\notin U} H^0(K_v,\widehat{C})\to H^1_{\mathrm{c}}(U,\widehat{\mathscr{C}})\to D^1(U,\widehat{\mathscr{C}})\to 0$$

is also exact by definition of $D^1(U,\widehat{\mathscr{C}})$. Using Lemma 3.11 and the fact that the ℓ-adic completion functor $^{(\ell)}$ commutes with finite direct sums, the sequence

$$\bigoplus_{v\notin U} H^0(K_v,\widehat{C})^{(\ell)}\to H^1_{\mathrm{c}}(U,\widehat{\mathscr{C}})^{(\ell)}\to D^1(U,\widehat{\mathscr{C}})^{(\ell)}\to 0$$

of profinite groups is exact as well, and its dual sequence (which is the second line of the diagram) remains exact.

The commutative diagram (34) defines a map

$$\theta: D^1(U,\mathscr{C})\{\ell\}\to (D^1(U,\widehat{\mathscr{C}})^{(\ell)})^*.$$

We observe that by [7, Th. 3.1] (which is the local duality theorem), the map β is an isomorphism, and we also know by Theorem 4.9 that ψ is an isomorphism. By diagram chasing θ is an isomorphism. Since this holds for every prime ℓ (including $\ell=p$), the proposition is proved, the finiteness of $D^1(U,\mathscr{C})$ and $D^1(U,\widehat{\mathscr{C}})$ being known by Proposition 3.6 (c). □

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